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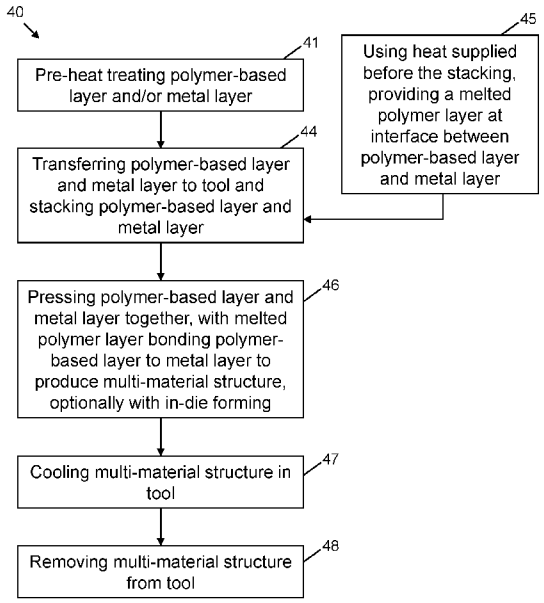


Figure 4

(57) Abstract: A method of manufacturing a multi-material structure (e.g. a fibre metal laminate) comprising a polymer-based layer (e.g. a fibre-reinforced polymer or fibre-reinforced thermoplastic) bonded to a metal layer, the method comprising: obtaining a polymer-based layer and a metal layer separate from one another; transferring the polymer-based layer and the metal layer to a tool and stacking the polymer-based layer and the metal layer, one on top of the other; using heat supplied before the stacking, providing a melted polymer layer at an interface between the polymer-based layer and the metal layer; and pressing together the polymer-based layer and the metal layer in the tool and allowing the melted polymer layer to cool and solidify; wherein the melted polymer layer bonds the polymer- based layer to the metal layer as it cools and solidifies, to produce the multi-material structure. Also provided is a multi-material structure manufactured by such a method.

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METHOD OF MANUFACTURING MULTI-MATERIAL STRUCTURES

Field of the Invention

The present invention relates to a method of manufacturing a multi-material structure comprising a polymer-based layer bonded to a metal layer. The invention is particularly applicable, but by no means limited, to manufacturing fibre metal laminate structures or components. Moreover, the method is particularly applicable, but by no means limited, to one-shot in-die forming of the multi-material structure – i.e. to imparting a contoured shape to the multi-material structure at the same time as the multi-material structure itself is manufactured.

Background of the Invention

A major trend in the transport industry is to improve energy efficiency by using lightweight materials. To this end, fibre metal laminates (FMLs) are a class of multi-material structures that are currently of increased interest. The term “fibre metal laminate” (or FML) as used herein refers to a multi-material structure comprising one or more metal layers bonded to one or more fibre-reinforced polymer (FRP) layers to form a laminate structure. The metal layer(s) may often comprise an alloy, although this need not be the case, as an unalloyed metal may be used instead.

Combining the merits of both metals and FRPs, FMLs have superior advantages including high strength-to-weight ratio, fatigue resistance and impact resistance. With these advantages, FMLs have become one of the state-of-the-art lightweight structural materials and have been successfully applied in the aviation and automotive industries, amongst others.

For example, in the aviation industry, GLARE (glass reinforced aluminium laminate), based on high-strength glass fibres, has been applied in several aircraft as the fuselage, wing panel and stringer materials, achieving a notable weight reduction of 30% compared to monometallic structures. Meanwhile, in the automotive industry, several automobile manufacturers have been using “FML-like” structures or FML components for the “Body-in-White” or “BiW” – this being the stage of manufacture when a car body’s frame has been joined together, before painting, and before the motor, chassis sub-assemblies, or trim (glass, seats, upholstery, door locks/handles, electronics, etc.) have been integrated into the structure.

"FML-like" structures are constructed by joining metal and FRP subcomponents using fasteners, adhesive bonding, or other joining technologies. For example, a carbon fibre-reinforced polymer (CFRP) reinforcement panel may be joined to an aluminium BiW, or a glass fibre-reinforced polymer (GFRP) may be joined to a steel component.

In contrast, an FML component is an integrally formed component consisting of metal and FRP layers without additional joining. Compared to FML-like structures, FML components have discernible advantages including better structural efficiency.

However, the application of FML components is limited by existing forming technologies. Currently, FML forming technologies fall into two major categories: one-shot forming and separate forming.

In one-shot forming technologies, it has been observed that the forming temperatures used for FMLs greatly coincide with those for FRPs based on the same thermoplastic matrices. This is to ensure the optimal formability of the FRP layers and not to damage the polymer matrix by excessive temperature during the forming process. Therefore, existing one-shot forming technology cannot be adopted to produce ultra-high-strength-steel (UHSS)-based FML components, for example, due to the mismatch between the preferred forming temperature of UHSSs and the preferred forming temperature of FRPs. In particular, the preferable forming temperatures of UHSSs for optimal formability and post-form structural performance are usually significantly higher than the critical temperature allowed for forming FRPs. For example, the forming temperature range of boron steel (22MnB5), a UHSS commonly used in BiW, is normally 700~850°C. Therefore, traditionally, UHSS-based FML components require separate forming of the FRP and metal layers – for example with CFRP prepregs and adhesive films being press formed onto a hot-stamped UHSS part, which was hot stamped in a separate step.

Separate forming technologies, especially when designed with hot-stamped UHSSs, offer more design flexibility and the ability to produce components with ultra-high performances. However, these methods require multiple sets of tools and involve complex multi-step processes. This leads to higher tooling costs, and higher manufacturing costs due to longer production time. In addition, separate forming often necessitates the use of adhesives to bond the metal and FRP layers, hindering the recyclability of the as-formed components.

Furthermore, for both one-shot and separate forming technologies, the stamped components needed to be held in pressurised tools for a certain amount of time, from minutes to hours, for the FRP to be consolidated or cured, depending on the polymer matrix and process design. This further increases the cycle time and thus manufacturing cost.

The present invention seeks to address at least some of the above issues.

Summary of the Invention

Aspects and embodiments of the present invention are set out in the appended claims.

According to a first aspect of the invention there is provided a method of manufacturing a multi-material structure (such as, but not limited to, an FML) comprising a polymer-based layer bonded to a metal layer, the method comprising: obtaining a polymer-based layer and a metal layer separate from one another; transferring the polymer-based layer and the metal layer to a tool and stacking the polymer-based layer and the metal layer, one on top of the other; using heat supplied before the stacking, providing a melted polymer layer at an interface between the polymer-based layer and the metal layer; and pressing together the polymer-based layer and the metal layer in the tool and allowing the melted polymer layer to cool and solidify; wherein the melted polymer layer bonds the polymer-based layer to the metal layer as it cools and solidifies, to produce the multi-material structure.

The term “tool” should be interpreted broadly, to encompass any apparatus in which pressure is applied to the polymer-based layer and the metal layer to press them together. For example, the tool may comprise a set of flat plates that can be brought together, or a set of shaped tools, e.g. comprising die, punch, blank holders and spacers, for in-die forming or stamp-forming a contoured part. The term “tool” also extends to apparatus used for electromagnetic forming or hydroforming components, for example.

It should also be understood that the term “melted”, in the context of providing a melted polymer layer, means that the polymer layer is heated to a temperature above its melting temperature, such that the polymer layer adopts a viscous, tacky or sticky state. There is no requirement for the polymer layer to transition fully into a molten liquid state, although this may also occur if desired.

The stacking may take place as the polymer-based layer and the metal layer are introduced into the tool, or beforehand.

By virtue of using heat supplied before the stacking to provide a melted polymer layer at an interface between the polymer-based layer and the metal layer, and using the subsequent solidification of that melted polymer layer to bond the polymer-based layer to the metal layer in the tool, the present work provides a versatile, efficient and effective way of manufacturing multi-material structures. In particular, it enables multi-material components to be formed in one-shot, offering significant advantages compared to existing technologies. The investment is minimised as only one set of tools and press equipment is required. The manufacturing cost is low due to one-shot forming and short in-die holding time. Moreover, no extra adhesives are required as the bonding is carried out by the thermoplastic of the polymer-based layer (e.g. a pure polymer sheet or fibre-reinforced polymer sheet), leading to better recyclability.

The present technique can be applied to a wide range of metal and polymer-based material combinations. Notably, for example, it allows one-shot forming of UHSS-based FMLs, which conventionally can only be produced by separate forming technologies due to the forming temperature mismatch.

The wide applicability of this technology offers an opportunity of tailoring the material combination to suit different design objectives, including weight, performance and cost, and promoting a wider application of multi-material structures.

The polymer-based layer may typically comprise a thermoplastic. The polymer-based layer may comprise a thermoplastic matrix composite, comprising a thermoplastic matrix with a reinforcing phase therein. Alternatively, however, it may be an unreinforced polymer layer.

When reinforced, the polymer-based layer may comprise a fibre reinforced thermoplastic. However, other types of reinforcement, instead of fibres, are also possible within the thermoplastic matrix, such as reinforcing particles.

The metal layer may typically comprise an alloy. For example, the alloy may be a steel, such as a UHSS (e.g. 22MnB5), or a high-strength aluminium alloy (e.g. AA6082). Alternatively, the metal layer may comprise an unalloyed metal, such as aluminium or

titanium, for example.

The resulting multi-material structure may comprise a fibre metal laminate.

The polymer-based layer may have a thickness in the range of 0.05 mm to 5 mm.

The metal layer may have a thickness in the range of 0.05 mm to 5 mm.

The tool may comprise a die (and e.g. a punch, a blank holder and spacers), and the pressing may further comprise performing in-die forming of the multi-material structure, to form a shaped/contoured structure or component.

Generally (but not in all cases necessary) the method may further comprise heating the polymer-based layer before transferring and stacking the polymer-based layer. The polymer-based layer may be heated to a temperature below the melting temperature of its constituent polymer. Alternatively, in other instances, the polymer-based layer may be pre-heat treated in other ways before transferring and stacking the polymer-based layer. The term “pre-heat treated” as used herein should be interpreted broadly, to encompass pre-heating and/or pre-cooling. This includes the possibility of a thermal profile that includes one or more heating stages followed by, preceded by, or interspersed by, one or more cooling stages. Thus, for instance, the polymer-based layer may be pre-heated to a temperature above the melting temperature of its constituent polymer, and may then be pre-cooled to achieve a desired temperature distribution before the stacking and, where applicable, forming. More generally, the polymer-based layer may be pre-heat treated to a temperature lower than, equal to, or higher than the melting temperature of its constituent polymer.

In a first example, the polymer layer that is melted is a surface layer of the polymer-based layer.

In the first example, the method may further comprise pre-heat treating the metal layer before stacking the metal layer. The metal layer may be pre-heat treated to a forming temperature higher than the melting temperature of the polymer. In some instances the metal layer may first be heated to an elevated temperature (e.g. an austenitisation temperature for steels, or a solution heat treatment temperature for aluminium alloys) higher

than the forming temperature, and soaked at the elevated temperature to impart a desired microstructure, and then the metal layer may be rapidly pre-cooled to the forming temperature, so as to retain the desired microstructure. Alternatively, the metal layer may be pre-heated directly to its forming temperature.

With the first example, the polymer layer may be melted by heat transferred from the metal layer to the polymer-based layer when the polymer-based layer and the metal layer are stacked, one on top of the other. This provides a way of selectively melting the surface of the polymer-based layer, to enable bonding with the metal layer.

In the first example, the metal layer may comprise a hot-formable alloy, such as a UHSS, for example.

In a second example, the polymer layer that is melted is again a surface layer of the polymer-based layer, but in this case, in the heating, additional heating is applied to the surface of the polymer-based layer, to melt the polymer layer. Again, this provides a way of selectively melting the surface of the polymer-based layer, to enable bonding with the metal layer. The additional heating may comprise radiation heating, convection heating, infrared heating, laser heating, electromagnetic induction heating, or contact heating, for example.

In the second example, the metal layer may comprise a metal that does not require being hot formed.

In third and fourth examples, the polymer layer that is melted is a polymer-based sheet or film initially separate from the polymer-based layer. The polymer-based sheet or film may comprise the same polymer as the polymer-based layer. Alternatively, though, the polymer-based sheet or film may comprise a different polymer from that of the polymer-based layer.

In the third example, the polymer-based sheet or film is heated to a temperature above the melting temperature of the polymer-based sheet or film while separate from the polymer-based layer and the metal layer, the polymer-based sheet or film is transferred separately from the polymer-based layer and the metal layer, and the polymer-based sheet or film is stacked between the polymer-based layer and the metal layer. In certain implementations, the metal layer is not heated.

In the fourth example, the polymer-based sheet or film is first placed on and heated together with the metal layer, to a temperature above the melting temperature of the polymer-based sheet or film, and is then transferred together with the metal layer. The metal layer may be pre-heat treated to a forming temperature before being stacked.

The polymer-based layer may also be heated to a temperature higher than the melting temperature of its constituent polymer. For instance, in a fifth example, the polymer layer that is melted is a surface layer of the polymer-based layer, and, in the heating, the entire polymer-based layer is heated to a temperature above the melting temperature of the polymer.

Optionally the metal layer may be pre-heat treated to a forming temperature before being stacked. For example, the metal layer may first be heated to an elevated temperature (e.g. an austenitisation temperature for steels, or a solution heat treatment temperature for aluminium alloys) higher than the forming temperature, and soaked at the elevated temperature to impart a desired microstructure, and then the metal layer may be rapidly pre-cooled to the forming temperature, so as to retain the desired microstructure. Alternatively, the metal layer may be pre-heated directly to its forming temperature, or may not be pre-heated.

The metal layer may comprise a hot-formable alloy, such as a UHSS.

In some other examples, the metal layer is not heated before being stacked.

With regard to all the aforementioned examples, heating or pre-heat treating of the polymer-based layer and/or the metal layer before being stacked may comprise furnace heating, radiation heating, convection heating, infrared heating, or electromagnetic induction heating, as well as convection- or contact-based pre-cooling, for instance.

The tool may be maintained at a temperature lower than the glass transition temperature of the polymer of the polymer-based layer.

The resulting multi-material structure may be cooled in the tool until it reaches a temperature lower than the glass transition temperature of its constituent polymer(s). The multi-material

structure may be cooled in the tool for less than one minute, preferably less than 30 seconds, preferably around 10 seconds.

The method may further comprise applying thermal treating to the tool. For example, the thermal treating may be controlled using active cooling systems and/or heating elements.

The method may further comprise quenching the multi-material structure in the tool.

Optionally, the method may further comprise performing a post-form heat treatment on the as-formed multi-material structure.

Optionally, the multi-material structure may comprise a plurality of polymer-based layers and/or a plurality of metal layers that are bonded by the method.

According to a second aspect of the invention there is provided a multi-material structure manufactured by the method of the first aspect.

Brief Description of the Drawings

Embodiments of the invention will now be described by way of example only with reference to the attached figures in which:

Figure 1 is a schematic cross-sectional illustration of part of a polymer-based layer and a metal layer, from which a multi-material structure is to be manufactured;

Figure 2 is a schematic cross-sectional illustration of a multi-material structure (in this case, a flat plate) comprising the polymer-based layer and metal layer of Figure 1, bonded together;

Figure 3 is a schematic cross-sectional illustration of another multi-material structure comprising a polymer-based layer bonded to a metal layer and stamped to produce a shaped multi-material component by one-shot forming;

Figure 4 is a flow diagram illustrating, in a general sense, a method of manufacturing a multi-material structure such as that of Figures 2 or 3;

Figure 5 is a flow diagram illustrating a first example method of manufacturing a multi-material structure such as that of Figures 2 or 3;

Figures 6a to 6e illustrate the process of bonding together a polymer-based layer and a metal layer in accordance with the first example method of Figure 5 to manufacture a flat multi-material structure;

Figures 7a to 7c illustrate the process of bonding together a polymer-based layer and a metal layer in accordance with the first example method of Figure 5 to manufacture a shaped multi-material structure by one-shot forming;

Figure 8 is a flow diagram illustrating a second example method of manufacturing a multi-material structure such as that of Figures 2 or 3;

Figures 9a to 9e illustrate the process of bonding together a polymer-based layer and a metal layer in accordance with the second example method of Figure 8 to manufacture a multi-material structure;

Figure 10 is a flow diagram illustrating third and fourth example methods of manufacturing a multi-material structure such as that of Figures 2 or 3;

Figures 11a to 11e illustrate the process of bonding together a polymer-based layer and a metal layer in accordance with the third example method of Figure 10 to manufacture a multi-material structure;

Figure 11f is a substitute for Figure 11a (Figures 11b to 11e then follow) to illustrate the process of bonding together a polymer-based layer and a metal layer in accordance with the fourth example method of Figure 10 to manufacture a multi-material structure;

Figure 12 is a flow diagram illustrating a fifth example method of manufacturing a multi-material structure such as that of Figures 2 or 3;

Figures 13a to 13e illustrate the process of bonding together a polymer-based layer and a metal layer in accordance with the fifth example method of Figure 12 to manufacture a multi-material structure;

Figure 14 is a photograph of an as-formed demonstrator component formed using the method of Figure 5, and including example dimensions in millimetres;

Figure 15 is a 3D-reconstructed photograph of computed tomography (CT) of the demonstrator component of Figure 14, viewed from above, and showing lines A-A, B-B and C-C through which cross-sectional CT scans were obtained;

Figure 16 shows the cross-sectional CT scans obtained along the lines A-A, B-B and C-C of Figure 15;

Figure 17 is a photograph of another as-formed demonstrator component formed using the method of Figure 5, and including example dimensions in millimetres; and

Figure 18 is an optical micrograph acquired at 50x magnification, showing a local view of the sidewall region of the demonstrator component of Figure 17.

In the figures, like elements are indicated by like reference numerals throughout.

Detailed Description

The present embodiments represent the best ways known to the Applicant of putting the invention into practice. However, they are not the only ways in which this can be achieved.

Multi-material structures that may be manufactured

With reference initially to Figures 1 to 3, the present method may be used to manufacture a wide range of multi-material structures comprising a polymer-based layer 12 bonded to a metal layer 14. As shown in Figure 1, the polymer-based layer 12 is initially separate from the metal layer 14; they are bonded together by the present method to produce a multi-material structure.

In some examples, as illustrated for instance in Figures 2 and 3, the multi-material structure 10, 20 may comprise a single polymer-based layer 12 bonded to a single metal layer 14, and the present description primarily focuses on such structures, for ease of understanding of the present principles. However, in other examples the multi-material structure may comprise a plurality of polymer-based layers and/or a plurality of metal layers, in any feasible layup arrangement, using the same bonding principles. In such cases, the polymer-based layers need not be identical with one another, and the metal layers need not be identical with one another. For example, the polymer-based layers may have different compositions and/or may have different thicknesses. Likewise, the metal layers may have different compositions and/or may have different thicknesses. A total of two or three constituent layers are generally preferred in the overall multi-material structure, as this is typically the limit of the automation capacity of a production line. However, more layers may be accommodated if the processing apparatus allows.

As shown in Figure 2, the resulting multi-material structure 10 may be plate-like, with the present technique simply bonding the polymer-based layer 12 to the metal layer 14. However, advantageously, the present technique may also be used to form a shaped or contoured multi-material structure 20, for example as illustrated in Figure 3, by one-shot in-die forming, with a shape being imparted to the multi-material structure at the same time as the multi-material structure itself is manufactured from its constituent layers.

Preferably the (or each) polymer-based layer 12 comprises a thermoplastic. Examples of suitable thermoplastics include, but are not limited to, a polyamide such as PA6 (or nylon 6), or a polyaryletherketone such as PEEK (polyether ether ketone).

The present work is particularly suitable to the production of multi-material structures in which the (or each) polymer-based layer 12 comprises a thermoplastic matrix composite. For example, the (or each) polymer-based layer may comprise a fibre reinforced polymer (FRP) or fibre reinforced thermoplastic (FRTP) having one or more constituent layers of fibres within a polymer matrix, in any feasible layup arrangement – although in other examples particles may be used instead of (or in addition to) fibres to provide reinforcement to the polymer matrix. When fibres are present within the polymer matrix, the fibres may be woven, unidirectional, or discontinuous, for example. Accordingly, the resulting multi-material structure may be a fibre metal laminate (FML) structure. As above, the polymer matrix within the FRP or FRTP layer(s) may comprise a thermoplastic such as (but not limited to) a polyamide such as PA6 (or nylon 6), or a polyaryletherketone such as PEEK.

Before beginning the present manufacturing method, the (or each) polymer-based layer 12 is typically obtained in prefabricated sheet form, typically having a thickness in the range of 0.05 mm to 5 mm.

The metal layer 14 may comprise an alloy, although unalloyed metals may be used instead. Examples of such alloys include steels, aluminium alloys and titanium alloys. The steel may be a UHSS (ultra-high-strength steel), such as 22MnB5, for instance. If unalloyed, the metal layer may be aluminium or titanium, for example.

Before beginning the present manufacturing method, the (or each) metal layer 14 is typically obtained in sheet form, typically having a thickness in the range of 0.05 mm to 5 mm.

The surface (i.e. plan view) dimensions of the polymer-based layer 12 may be the same as those of the metal layer 14. Alternatively, the surface (i.e. plan view) dimensions of the polymer-based layer 12 may be different from those of the metal layer 14. In other words, the polymer-based layer 12 and the metal layer 14 may be the same size, or different sizes. For example, the polymer-based layer 12 may take the form of a smaller patch applied to a region of a metal component, in order to provide local reinforcement in that region.

Method overview

Figure 4 is a flow diagram illustrating, in a general sense, the main principles of the present manufacturing method 40, depicted through a number of constituent blocks. The method

40 begins with the polymer-based layer 12 and the metal layer 14 being separate from one another. Commonly, but not necessarily always, in block 41 the polymer-based layer 12 and/or the metal layer 14 are pre-heat treated to a respective forming temperature, with such heating being carried out using separate heating apparatus(es) and/or cooling apparatus if desired. That is to say (and with reference in passing to Figure 6a), if the polymer-based layer 12 is to be heated, it may be heated in a dedicated heating apparatus 4, separate from the metal layer 14. Likewise, if the metal layer 14 is to be heated, it may be pre-heat treated in a dedicated heating apparatus(es) 6, separate from the polymer-based layer 12. Such pre-heat treating may be necessary if the multi-material structure is to be in-die formed into a shaped structure or component 20 during manufacture, as in Figure 3. Alternatively, instead of employing separate dedicated heating apparatuses for pre-heat treating the polymer-based layer 12 and the metal layer 14, a single heating apparatus may be used having separate heating profiles or zones for pre-heat treating the polymer-based layer 12 and the metal layer 14 respectively.

For all the presently-disclosed pre-heat treatment options, the heating apparatus 4 for heating the polymer-based layer 12, or the heating apparatus 6 for heating the metal layer 14, may be (but is not limited to) any suitable one of: furnaces, radiation heating apparatus, convection heating apparatus, infrared heating apparatus or induction heating apparatus. The cooling apparatus for pre-cooling the metal layer 14 or the polymer-based layer 12, if desired, may be (but it not limited to) any suitable one of: compressed air cooling, contact cooling.

With regard to the metal layer 14, the pre-heating may comprise heating the metal layer 14 to an elevated temperature higher than its forming temperature (e.g. above the austenitisation temperature for steels, or above the solution heat treatment temperature for aluminium alloys), soaking the metal layer at the elevated temperature to impart a desired microstructure, and then rapidly pre-cooling (e.g. at a rate of -30°C/s) the metal layer to the forming temperature, so as to retain the desired microstructure. The pre-cooling may take place before the metal layer is stacked with the polymer-based layer(s).

As depicted by block 44 of Figure 4, the method 40 then comprises transferring the polymer-based layer 12 and the metal layer 14 to a tool (see e.g. 16 in Figure 6b; or 17, 18 and 19 in Figure 7a) and stacking the polymer-based layer 12 and the metal layer 14, one on top of the other. The stacking may take place as the polymer-based layer 12 and the metal

layer 14 are introduced into the tool, or beforehand. The transfer is performed rapidly (i.e. in a few seconds or so), to avoid significant loss of heat from the layers to be joined. Typically the polymer-based layer 12 is stacked on top of the metal layer 14, but it is also possible in some circumstances for the metal layer 14 to be stacked on top of the polymer-based layer 12. When stacked, an interface is formed between the polymer-based layer 12 and the metal layer 14.

Once stacked, no additional heating is applied to the polymer-based layer 12 or the metal layer 14 to cause the polymer-based layer 12 to bond to the metal layer 14. In particular, the tool itself may not necessarily need to be heated.

Rather, as represented by block 45 of Figure 4, using heat supplied before the polymer-based layer 12 and the metal layer 14 are stacked, the method comprises providing a melted polymer layer 13 at the interface between the polymer-based layer 12 and the metal layer 14. As discussed in greater detail below, in some examples (e.g. the present Example 1; see e.g. Figure 6b) the melted polymer layer 13 may be a surface layer of the polymer-based layer 12, that is melted by heat (H in Figure 6b) transferred from the adjacent metal layer 14, due to the metal layer 14 having been pre-heat treated to a higher temperature than the polymer-based layer 12. In other examples (e.g. the present Example 2; see e.g. Figure 9a) the melted polymer layer 13 may be a surface layer of the polymer-based layer 12, that is melted by additional heating (SH in Figure 9a) applied to the surface of the polymer-based layer 12 during pre-heat treating. In other examples (e.g. the present Examples 3 and 4; see e.g. Figure 11a) the melted polymer layer 13 may be a separate polymer-based sheet or film that is heated and melted prior to being stacked, whereupon it is inserted between the polymer-based layer 12 and the metal layer 14. In other examples (e.g. the present Example 5; see e.g. Figure 13a) the polymer layer 13 that is melted is the overall polymer-based layer 12, that is pre-heated to above its melting temperature in its entirety prior to being stacked. (In general, it is noted in passing that the polymer layer 13 and the polymer-based layer 12 may or may not be at the same temperature.)

Then, as depicted by block 46 of Figure 4, the method comprises pressing together the polymer-based layer 12 and the metal layer 14 in the tool and allowing the melted polymer layer 13 to cool and solidify, with the melted polymer layer 13 bonding the polymer-based layer 12 to the metal layer 14 as the melted polymer layer 13 cools and solidifies. If required, the multi-material structure may be in-die formed into a desired shape (such as that of multi-

material component 20 in Figure 3) during the pressing process; although alternatively the multi-material structure may simply be of flat plate form, by being pressed between flat plates. Suitable techniques for carrying out in-die forming of a shaped multi-material component include, but are not limited to, stamping (using a matched set of tools, e.g. die and punch, as illustrated in Figures 7a-7b), electromagnetic forming, hydroforming, etc.

The multi-material structure is cooled (block 47) before being removed from the tool (block 48). The cooling process may occur passively, or alternatively dedicated cooling may be applied to the tool, e.g. using an active cooling system. Optionally the cooling may comprise in-die quenching. To reduce the holding time, cold tools are preferred to facilitate in-die quenching. To adjust the cooling rate of in-die quenching, the exact temperature of the tools may be tuned by active cooling systems or heating elements.

More particularly, the tools, including the dies and/or other tools that are in direct contact with the stacked polymer-based layer 12 and/or metal alloy sheets 14, are preferably at a temperature lower than the glass transition temperature (Tg) of the constituent polymer(s). The formed multi-material product 10, 20 is preferably held in the die set and cooling is performed for a period of time sufficiently long to cool the formed product 10, 20 to a temperature lower than the Tg of the constituent polymer(s). A typical cooling time may be below one minute, and is preferably below 30 seconds, preferably around 10 seconds.

The above manufacturing process may be followed by one or more post-form heat treatments, such as ageing, if required.

Expressed differently, the present technique involves setting up a potentially tailored through-the-thickness temperature field in the polymer-based layer 12 directly prior to forming, where the polymer surface 13 in contact with the metal layer 14 is melted and the rest of the polymer-based layer 12 can be either melted or solid-state. Upon cooling and solidifying, the melted polymer at the metal-polymer interface bonds the two materials (layers 12 and 14) together without requiring additional adhesive.

To cater for different material combinations used to manufacture the multi-material structure, and to take into account the materials' preferred forming temperatures, the through-the-thickness temperature field can be achieved by different pre-heat treatment routes, as set out in the examples below.

When the preferred forming temperature of the polymer-based layer 12 is below the melting temperature (Tm) of its constituent polymer(s), the polymer-based layer 12 may be heated to and soaked at a designated temperature below the melting point Tm before being transferred and stacked with the metal layer 14. When the preferred forming temperature of the polymer-based layer 12 is above the melting temperature (Tm) of its constituent polymer(s), the polymer-based layer 12 may be heated to and soaked at a designated temperature above the melting point Tm before being transferred and stacked with the metal layer 14.

Example 1

This example applies to metals or metal alloys which require being hot formed for optimal formability and post-form performance, such as UHSS 22MnB5 and aluminium alloy 6082.

In this process, the metal layer 14 is firstly heated and soaked at an elevated temperature, which can be above the austenitisation temperature for steels or the solution heat treatment temperature for aluminium alloys, to impart a desired microstructure. The metal layer 14 is then rapidly pre-cooled to a suitable forming temperature that is sufficiently low as to avoid causing damage or degradation to the polymer-based layer 12 during stacking, while still retaining the preferred imparted microstructure of the metal layer 14. The metal layer 14, which is still at a sufficiently high temperature after pre-cooling, higher than the melting temperature of the polymer, acts as a heat source to melt a polymer surface layer 13 of the polymer-based layer 12 during stacking.

More generally, the abovementioned pre-cooling step may be carried out on one or both of the metal layer 14 and the polymer-based layer 12, to bring the material system (both the metal and polymer-based layers) to a suitable temperature distribution for stacking and forming. The target temperature for the metal layer 14 is sufficiently low to avoid causing damage or degradation to the polymer-based layer 12 during stacking, while still retaining the preferred imparted microstructure of the metal layer 14. The metal layer 14 remains at a sufficiently high temperature after such pre-cooling, higher than the melting temperature of the polymer, to enable it to act as a heat source to melt a polymer surface layer 13 of the polymer-based layer 12 during stacking.

It should be noted the pre-cooling of the metal layer 14 or polymer-based layer 12 may be

optional if the metal layer 14, when heat treated to its preferred forming temperature, will not cause damage to the polymer-based layer 12 upon stacking.

In more detail, with reference to Figures 5 and 6a-6e, the method 50 of Example 1 comprises the following:

The metal layer 14 is pre-heated in heating apparatus 6, to a temperature higher than the melting temperature (Tm) of the polymer of the polymer-based layer 12 (see block 51 and Figure 6a). As noted above, the pre-heating may involve heating and soaking the metal layer 14 at an elevated temperature, followed by rapidly pre-cooling the metal layer 14 to a forming temperature that is still higher than the melting temperature of the polymer of the polymer-based layer 12, in order to impart a desired microstructure on the metal layer 14 and confer optimal formability.

In parallel, the polymer-based layer 12 is heated in separate heating apparatus 4, to a temperature lower than Tm of its constituent polymer (see block 52 and Figure 6a). As part of this, the polymer-based layer 12 may be subjected to pre-cooling (potentially rapid pre-cooling) after heating.

Then, as depicted by block 54, the polymer-based layer 12 and the metal layer 14 are rapidly transferred, stacked and placed in the tool 16 (see Figure 6b). Here, the metal layer 14 acts as a heat source and transfers heat H to the polymer-based layer 12. This causes a surface layer 13 of the polymer-based layer 12 to melt, at the interface between the polymer-based layer 12 and the metal layer 14, forming a melted polymer layer 13 (block 55).

The polymer-based layer 12 and the metal layer 14 are then pressed together (block 56 and Figure 6c) through the application of pressure P to the tool 16, with the melted polymer layer 13 bonding the polymer-based layer 12 to the metal layer 14 as the melted polymer layer 13 cools and solidifies. Once the melted polymer layer 13 has solidified, this results in the configuration shown in Figure 6d, in which the polymer-based layer 12 is bonded to the metal layer 14 within the tool 16, forming the multi-material structure 10.

Following cooling of the multi-material structure (block 57), the tool 16 is then opened (Figure 6e) and the multi-material structure 10 can then be removed (block 58).

If it is desired to shape the multi-material structure then, at block 56, in-die forming of the multi-material structure may be performed during the pressing process. Figures 7a-7b illustrate an example of this. Here, the tool comprises complementary die parts 17 and 19 and blank holder 18. When placed in the tool, the polymer-based layer 12 and the metal layer 14 are initially stacked on the blank holder 18. Again, the metal layer 14 acts as a heat source and transfers heat to the polymer-based layer 12, causing a surface layer 13 of the polymer-based layer 12 to melt. At this point, the tool is closed by bringing the die parts 17 and 19 together under pressure P, imparting a contoured shape on the polymer-based layer 12 and the metal layer 14 whilst pressing the polymer-based layer 12 and the metal layer 14 together. Once the melted polymer layer 13 has solidified, the polymer-based layer 12 and the metal layer 14 are bonded together, thereby forming the resulting multi-material structure 20 of Figure 7c, which can then be removed from the tool once sufficiently cool.

Example 2

This example applies to metals or metal alloys which do not require being hot formed, such as various types of coldwork steels, aluminium alloys, and titanium. Accordingly, in this example, the metal layer 14 can be formed at room temperature and need not be pre-heated.

Since the metal layer 14 is not heated (or is not heated sufficiently to act as a heat source to melt the surface of the polymer-based layer 12), in this example an additional heating step is provided to melt a surface layer 13 of the polymer-based layer 12. This additional surface heating (SH in Figure 9a) can be realised through a range of possible techniques, such as radiation heating, convection heating, infrared heating, laser heating, electromagnetic induction heating, or contact heating.

In more detail, with reference to Figures 8 and 9a-9e, the method 80 of Example 2 comprises the following:

The polymer-based layer 12 is heated, in dedicated heating apparatus 4, to a temperature lower than Tm of its constituent polymer (see block 81 and Figure 9a). As part of this, the polymer-based layer 12 may be subjected to pre-cooling (potentially rapid pre-cooling) after heating. In parallel, the metal layer 14 may be pre-heat treated if desired (see block 83), in

separate heating apparatus 6, to a forming temperature for optimal formability.

Notably, in block 82, additional heating (SH in Figure 9a) is applied to the surface 13 of the polymer-based layer 12, to form a melted polymer layer 13.

Then, as depicted by block 84, the polymer-based layer 12 and the metal layer 14 are rapidly transferred, stacked and placed in the tool 16 (see Figure 9b), with the melted polymer surface layer 13 at the interface between the polymer-based layer 12 and the metal layer 14.

Alternatively, or as well, the additional heating (SH) may take place during the transferring and/or the stacking.

The polymer-based layer 12 and the metal layer 14 are then pressed together (block 86 and Figure 9c) through the application of pressure P to the tool, with the melted polymer layer 13 bonding the polymer-based layer 12 to the metal layer 14 as the melted polymer layer 13 cools and solidifies. Once the melted polymer layer 13 has solidified, this results in the configuration shown in Figure 9d, in which the polymer-based layer 12 is bonded to the metal layer 14 within the tool 16, forming the multi-material structure 10.

Following cooling of the multi-material structure (block 87), the tool 16 is then opened (Figure 9e) and the multi-material structure 10 can then be removed (block 88).

If it is desired to shape the multi-material structure then, at block 86, in-die forming of the multi-material structure may be performed during the pressing process, for example in the manner illustrated in Figures 7a-7b.

Examples 3 and 4

These two examples are alternatives in which a separate supplementary layer of melted polymer-based sheet or film 13 is introduced between the polymer-based layer 12 and the metal layer 14. The supplementary polymer-based sheet or film 13 may for example be an FRTP sheet or a thermoplastic film; the constituent polymer of the sheet of film 13 may be the same or different material as the polymer of the polymer-based layer 12. Using the same polymer material in both the polymer-based layer 12 and the supplementary sheet or film 13 provides better recyclability. However, using different polymer materials in the

polymer-based layer 12 and the supplementary sheet or film 13 can also provide advantages. For example, the polymer of the polymer-based layer 12 could be PEEK, and the polymer of the supplementary polymer-based sheet or film 13 could be PA6. As the melting temperature of PA6 is much lower than that of PEEK, it can be heated to a relatively lower temperature to melt it, thereby reducing energy consumption whilst still effecting bonding between the polymer-based layer 12 and the metal layer 14.

Typically, but not necessarily, the thickness of the supplementary polymer-based sheet or film 13 is less than the thickness of the primary polymer-based layer 12 and the metal layer 14.

In Example 3, the polymer-based sheet or film 13 is heated to above its Tm and transferred separately from the polymer-based layer 12 and the metal layer 14. In Example 4, the polymer-based sheet or film 13 is heated and transferred together with the metal layer 14.

In more detail, with reference to Figures 10 and 11a-11f, the method 100 of Examples 3 and 4 comprises the following:

In Example 3 the polymer-based layer 12 is heated, in dedicated heating apparatus 4, to a temperature lower than Tm of its constituent polymer (see block 101 and Figure 11a). In parallel, the metal layer 14 may be pre-heat treated if desired, in separate heating apparatus 6, to a preferred forming temperature for optimal formability. Also in parallel, in block 102, a separate polymer-based sheet or film 13 (separate from the polymer-based layer 12 and the metal layer 14) is heated in separate heating apparatus 5 to above its melting temperature, to form a melted polymer layer.

On the other hand, in Example 4 the polymer-based layer 12 is heated, in dedicated heating apparatus 4, to a temperature lower than Tm of its constituent polymer (see block 101 and Figure 11f). In parallel, in block 102, the metal layer 14, with the polymer-based sheet or film 13 placed adjacently (e.g. on top or underneath), is heated in separate heating apparatus 6 to a temperature above the melting temperature of the polymer-based sheet or film 13. If appropriate, the temperature may be higher still, to reach or approach a preferred forming temperature of the metal layer 14.

Then, in both Example 3 and Example 4, and as depicted by block 104, the polymer-based

layer 12 and the metal layer 14 are rapidly transferred, stacked and placed in the tool 16 (see Figure 11b), with the melted polymer-based sheet or film 13 in-between, at the interface between the polymer-based layer 12 and the metal layer 14. In the case of Example 3, the melted polymer-based sheet or film 13 is transferred separately from the polymer-based layer 12 and the metal layer 14. In the case of Example 4, the melted polymer-based sheet or film 13 is transferred together with (i.e. on) the metal layer 14.

The polymer-based layer 12 and the metal layer 14 are then pressed together (block 106 and Figure 11c) through the application of pressure P to the tool, with the melted polymer-based sheet or film 13 bonding the polymer-based layer 12 to the metal layer 14 as the melted polymer-based sheet or film 13 cools and solidifies. Once the melted polymer-based sheet or film 13 has solidified, this results in the configuration shown in Figure 11d, in which the polymer-based layer 12 is bonded to the metal layer 14 within the tool 16, forming the multi-material structure 10.

Following cooling of the multi-material structure (block 107), the tool 16 is then opened (Figure 11e) and the multi-material structure 10 can then be removed (block 108).

If it is desired to shape the multi-material structure then, at block 106, in-die forming of the multi-material structure may be performed during the pressing process, for example in the manner illustrated in Figures 7a-7b.

Example 5

In this example, when the preferred forming temperature of the polymer-based layer 12 is above the Tm of the constituent polymer, the polymer-based layer 12 is heated directly and uniformly to a temperature above its Tm. Accordingly, the entirety of the polymer-based layer 12 is melted, thus providing a melted polymer layer 13 at the surface of the polymer-based layer 12, for bonding to the metal layer 14 in the tool. Therefore, unlike Examples 1-4, additional steps for creating the melted polymer layer 13 are not needed.

Example 5 may be applied to metals or metal alloys which require being hot formed for optimal formability and post-form performance, with the metal layer 14 being pre-heat treated to a forming temperature before being transferred. For example, a pre-heat treatment process similar to that described in Example 1 may be applied (i.e. to an elevated temperature to achieve austenitisation or solution heat treatment, followed by a rapid pre-

cooling step to bring the material system (both the metal and polymer-based layers) to the forming temperatures if necessary). Similar to Examples 2-4, the metal layer 14 can optionally be heated to a preferred forming temperatures for optimal formability. With respect to Example 5 it should be noted that, although the through-the-thickness temperature field is not tailored for the polymer-based layer 12, a temperature gradient may still be introduced in the polymer-based layer 12 prior to forming, due to different stacking temperatures of the polymer-based layer 12 and the metal layer 14.

In more detail, with reference to Figures 12 and 13a-13e, the method 120 of Example 5 comprises the following:

The polymer-based layer 12 is heated, in dedicated heating apparatus 4, to a temperature higher than Tm of its constituent polymer (see block 121 and Figure 13a). In parallel, the metal layer 14 may be pre-heat treated if desired, in separate heating apparatus 6, to a preferred forming temperature for optimal formability (see block 122 and Figure 13a).

Then, as depicted by block 124, the melted polymer-based layer 12 and the metal layer 14 are rapidly transferred, stacked and placed in the tool 16 (see Figure 13b), notionally with the melted polymer surface layer 13 at the interface with the metal layer 14.

The polymer-based layer 12 and the metal layer 14 are then pressed together (block 126 and Figure 13c) through the application of pressure P to the tool, with the melted polymer layer 13 bonding the polymer-based layer 12 to the metal layer 14 as the melted polymer layer 13 (and indeed the entire polymer-based layer 12) cools and solidifies. Once the melted polymer layer 13 has solidified, this results in the configuration shown in Figure 13d, in which the polymer-based layer 12 is bonded to the metal layer 14 within the tool 16, forming the multi-material structure 10.

Following cooling of the multi-material structure (block 127), the tool 16 is then opened (Figure 13e) and the multi-material structure 10 can then be removed (block 128).

If it is desired to shape the multi-material structure then, at block 126, in-die forming of the multi-material structure may be performed during the pressing process, for example in the manner illustrated in Figures 7a-7b.

A variant of Example 5 involves no heating of the metal layer 14 before the layers 12, 14 are transferred and stacked. That is to say, block 122 of Figure 12 is omitted. Accordingly, this variant may be applied to metals and metal alloys which do not need to be hot formed.

Summary of pre-heat treatment methods of Examples 1-5

The following is a summary of the pre-heat treatment methods employed in Examples 1-5 above.

Example 1: (i) Pre-heat the polymer-based layer 12 to a temperature lower than Tm; (ii) pre-heat the metal layer 14, followed by pre-cooling to a suitable temperature (higher than the Tm of the polymer-based layer); (iii) transfer and stack the polymer-based layer 12 with the metal layer 14; (iv) melt the surface 13 of the polymer-based layer 12 by interfacial heat transfer. This method is preferred when the metal layer is a hot-stamping advanced high-strength steel, for example.

Example 2: (i) Pre-heat the polymer-based layer 12 to a temperature lower than Tm; (ii) melt the surface 13 of the polymer-based layer 12 by additional surface heating; (iii) transfer and stack the polymer-based layer 12 with the metal layer 14.

Example 3: (i) Pre-heat the primary polymer-based layer 12 to a temperature lower than Tm; (ii) pre-heat a separate supplementary polymer-based sheet or film 13 to above Tm; (iii) transfer and stack the polymer-based layer 12 and the metal layer 14, with the supplementary polymer-based sheet or film 13 between the primary polymer-based layer 12 and the metal layer 14.

Example 4: (i) Pre-heat the primary polymer-based layer 12 to a temperature lower than Tm; (ii) stack a separate supplementary polymer-based sheet or film 13 with the metal layer 14; (iii) pre-heat the stacked supplementary polymer-based sheet or film 13 and the metal layer 14 to a temperature above the Tm of the supplementary polymer-based sheet or film 13; (iv) transfer and stack the primary polymer-based layer 12 and the metal layer 14, with the supplementary polymer-based sheet or film 13 between the primary polymer-based layer 12 and the metal layer 14.

Example 5: (i) Pre-heat the polymer-based layer 12 to a temperature above Tm; (ii) transfer and stack the polymer-based layer 12 with the metal layer 14.

In Examples 2, 3 and 5, the metal layer 14 can undergo any suitable temperature profile. Examples include:

a. Keep at room temperature.

b. Pre-heat to a suitable forming temperature.

c. Pre-heat to a suitable elevated temperature, followed by pre-cooling to a suitable stacking or forming temperature.

In all of Examples 1-5, the polymer-based layer 12 may undergo pre-cooling after heating and prior to stacking or forming.

Advantages

The presently-disclosed techniques have wide applicability for producing FML components (or other multi-material structures) of different performance grades, including ultra-high performance by integrating hot stamping of (ultra-)high strength alloys, for various structural application requirements. As a one-shot forming technology requiring only one set of tools and press equipment, the present techniques are cost-effective to implement, require minimal investment, and can be applied to existing (hot) stamping production lines as an upgrade.

Using the present techniques, product manufacturing cost can be reduced due to significantly shortened cycle time compared to existing one-shot or separate forming technologies. Moreover, as adhesive-free technology, the present techniques provide enhanced recyclability compared to separate forming technologies. Table 1 below summarises the advantages of the present work compared to different existing technologies, especially stamping technologies for producing FML components as detailed below.

Existing technologies	**Advantages of the presently-disclosed forming techniques**
Stamping	
One-shot thermoforming (heating metal and FRP together) - FRP matrix: mostly thermoplastic (PP, PA6, PEEK)	- Product performance: capable of producing (ultra-)high strength FML components by integrating hot stamping (ultra-)high strength alloys

- Metal: steels, aluminium alloys - Forming FMLs in laminates or stacks - Hot tools	- Cycle time <5% (compared to a common consolidation time of ~20 min) - Tooling cost saving ~50%
One-shot thermoforming (separate heating) - FRP matrix: thermoset (epoxy), thermoplastic (PA6) - Metal: steels - Forming FMLs in stack - Hot tools	- Product performance: capable of producing (ultra-)high strength FML components by integrating hot stamping (ultra-)high strength alloys - Cycle time <2% (compared to a curing time of ~45 mins) - Tooling cost saving ~50%
Separate forming - FRP matrix: thermoset (epoxy) - Metal: steels, including UHSS - Press FRP prepregs (adhesive coated) onto hot-stamped steel component	- Product performance: equivalent - Manufacturing cost saving >70% - Tooling cost saving >50% - Better recyclability with thermoplastic and without adhesives
Other forming methods	
Layup/shot peening forming/laser forming	Significantly reduced cycle time, more complex component geometry compared to large single/shallow double curvature panel components
Press brake bending	More complex component geometry compared to single curvature bending
Incremental sheet forming	Significantly reduced cycle time, lower equipment cost

Table 1: Summary of advantages of the proposed methods

Practical demonstration 1 – UHSS and CF/PEEK

To demonstrate the present technique (specifically the method according to Example 1), a shaped multi-material component was produced using hot stamping and in-die quenching. Here, the metal layer 14 was a UHSS (specifically 22MnB5) and the polymer-based layer 12 was an FRP comprising carbon fibres in a PEEK matrix (abbreviated herein to "CF/PEEK"). The method of Example 1 was employed for this demonstration since this is the preferred method when the metal layers are hot-stamping high strength metal alloys,

such as ultra-high strength steels (UHSSs). The forming apparatus and process were as illustrated in Figures 7a, 7b and 7c.

In this demonstration, the metal alloy UHSS 22MnB5 was heated and soaked at an elevated temperature to achieve austenitisation, then pre-cooled to a suitable forming temperature that was still sufficiently high to melt the surface of the CF/PEEK, thereby enabling post-form bonding while avoiding degradation. The overall forming process was as follows:

Step 1 (heating and soaking): heat the 22MnB5 (layer 14) and soak for complete austenitisation; heat CF/PEEK (layer 12) to and soak at a temperature between Tg and Tm.

Step 2 (pre-cooling): transfer the 22MnB5 and position on the tool (blank holder 18 of Figure 7a), followed by active air pre-cooling.

Step 3 (stacking): transfer the CF/PEEK and stack it on the pre-positioned 22MnB5.

Step 4 (stamping): close and hold the cold tools for stamping and in-die quenching.

Step 5 (removal): remove the formed component (structure 20) when the temperature is below the glass transition temperature (Tg) of PEEK.

A demonstrator structure 20 as shown in Figure 14 was successfully formed using one layer of 1 mm thick 22MnB5, and one layer of 2.15 mm thick CF/PEEK, using the process and process parameters detailed above. The bonding quality between the 22MnB5 and CF/PEEK interface was examined by computed tomography (CT) along the lines A-A, B-B and C-C indicated in Figure 15. Figure 16 shows three cross sections from the CT scan, demonstrating that good bonding quality was successfully and uniformly attained.

Practical demonstration 2 – Aluminium Alloy and CF/PA6

To further illustrate the applicability of the disclosed method (according to Example 1) beyond UHSS 22MnB5 and CF/PEEK, a second multi-material demonstrator component was produced. Here the metal layer 14 was an aluminium alloy (specifically AA6082) and the polymer-based layer 12 was an FRP comprising carbon fibres in a PA6 matrix (abbreviated herein to “CF/PA6”). The method of Example 1 was again employed for this aluminium-based demonstration, as the same forming sequence is suitable for hot-stamping aluminium alloys such as AA6082, which are commonly heated to elevated temperatures for solution heat treatment before forming. The forming apparatus and process were as illustrated in Figures 7a, 7b and 7c.

In this demonstration, the aluminium alloy AA6082 was heated and soaked at an elevated

temperature suitable for its solution heat treatment, then transferred directly to the forming tool at a temperature that was sufficiently high to melt the surface of the CF/PA6, thereby enabling interfacial bonding while avoiding degradation of the composite. The overall forming process proceeded as follows:

Step 1 (heating and soaking): heat the AA6082 sheet (layer 14) and soak at an elevated temperature suitable for its solution heat treatment; heat the CF/PA6 (layer 12) to and soak at a temperature between Tg and Tm.

Step 2 (aluminium transfer): Transfer the AA6082 and position it on the tool (blank holder 18 of Figure 7a).

Step 3 (stacking): transfer the CF/PA6 and stack it on the pre-positioned AA6082.

Step 4 (stamping): close and hold the cold tools for stamping and in-die quenching.

Step 5 (removal): remove the formed component (structure 20) when the temperature is below the glass transition temperature (Tg) of PA6.

Step 6 (aging and paint bake): subject the removed component (structure 20) to an artificial-aging treatment (190°C for 8 hours) followed by a paint-bake cycle (180°C for 30 minutes).

A demonstrator structure 20 as shown in Figure 17 was successfully formed using one layer of 1.2 mm thick AA6082, and one layer of 2 mm thick CF/PA6, using the process and process parameters detailed above. The bonding quality between the AA6082 and CF/PA6 interface was examined by optical microscopy. Figure 18 is an optical micrograph acquired at 50x magnification, showing a local view of the sidewall region of the demonstrator component of Figure 17, and demonstrating that good bonding quality was successfully attained.

Modifications and alternatives

Detailed embodiments and some possible alternatives have been described above. As those skilled in the art will appreciate, a number of modifications and further alternatives can be made to the above embodiments whilst still benefiting from the inventions embodied therein.

Each feature disclosed in the description, and (where appropriate) the claims and drawings may be provided independently or in any appropriate combination.

Reference numerals appearing in the claims are by way of illustration only and shall have

no limiting effect on the scope of the claims.

CLAIMS

1. A method of manufacturing a multi-material structure comprising a polymer-based layer bonded to a metal layer, the method comprising:

obtaining a polymer-based layer and a metal layer separate from one another;

transferring the polymer-based layer and the metal layer to a tool and stacking the polymer-based layer and the metal layer, one on top of the other;

using heat supplied before the stacking, providing a melted polymer layer at an interface between the polymer-based layer and the metal layer; and

pressing together the polymer-based layer and the metal layer in the tool and allowing the melted polymer layer to cool and solidify;

wherein the melted polymer layer bonds the polymer-based layer to the metal layer as it cools and solidifies, to produce the multi-material structure.

2. The method according to claim 1, wherein the polymer-based layer comprises a thermoplastic.

3. The method according to claim 2, wherein the polymer-based layer comprises a thermoplastic matrix composite.

4. The method according to claim 3, wherein the polymer-based layer comprises a fibre reinforced thermoplastic.

5. The method according to any preceding claim, wherein the metal layer comprises an alloy.

6. The method according to any preceding claim, wherein the multi-material structure comprises a fibre metal laminate.

7. The method according to any preceding claim, wherein the polymer-based layer has a thickness in the range of 0.05 mm to 5 mm.

8. The method according to any preceding claim, wherein the metal layer has a thickness in the range of 0.05 mm to 5 mm.

9. The method according to any preceding claim, wherein the tool comprises a die, and wherein the pressing further comprises performing in-die forming of the multi-material structure.

10. The method according to any preceding claim, further comprising heating the polymer-based layer before the transferring and stacking.

11. The method according to claim 10, wherein the polymer-based layer is heated to a temperature below the melting temperature of its constituent polymer.

12. The method according to any preceding claim, wherein the melted polymer layer is a surface layer of the polymer-based layer.

13. The method according to claim 12, further comprising pre-heat treating the metal layer before stacking the metal layer.

14. The method according to claim 13, wherein the metal layer is pre-heat treated to a forming temperature higher than the melting temperature of the polymer.

15. The method according to claim 14, wherein the metal layer is first heated to an elevated temperature higher than the forming temperature, and then is rapidly pre-cooled to the forming temperature.

16. The method according to any of claims 13 to 15, wherein the polymer layer is melted by heat transferred from the metal layer to the polymer-based layer when the polymer-based layer and the metal layer are stacked, one on top of the other.

17. The method according to any of claims 13 to 16, wherein the metal layer comprises a hot-formable alloy.

18. The method according to claim 12 when dependent on claim 11, wherein, in the heating, additional heating is applied to the surface of the polymer-based layer, to melt the polymer layer.

19. The method according to claim 18, wherein the additional heating comprises

radiation heating, convection heating, infrared heating, laser heating, electromagnetic induction heating, or contact heating.

20. The method according to claim 18 or claim 19, wherein the metal layer does not require being hot formed.

21. The method according to any of claims 1 to 11, wherein the polymer layer is a polymer-based sheet or film initially separate from the polymer-based layer.

22. The method according to claim 21, wherein the polymer-based sheet or film comprises the same polymer as the polymer-based layer.

23. The method according to claim 21, wherein the polymer-based sheet or film comprises a different polymer from that of the polymer-based layer.

24. The method according to claim 22 or claim 23, wherein the polymer-based sheet or film is heated to a temperature above the melting temperature of the polymer-based sheet or film while separate from the polymer-based layer and the metal layer, wherein the polymer-based sheet or film is transferred separately from the polymer-based layer and the metal layer, and wherein the polymer-based sheet or film is stacked between the polymer-based layer and the metal layer.

25. The method according to claim 24, wherein the metal layer is not heated.

26. The method according to claim 22 or claim 23, wherein the polymer-based sheet or film is first placed on and heated together with the metal layer, to a temperature above the melting temperature of the polymer-based sheet or film, and is then transferred together with the metal layer.

27. The method according to any of claims 21 to 24 or claim 26, wherein the metal layer is pre-heat treated to a forming temperature before being stacked.

28. The method according to claim 10, wherein the polymer layer is a surface layer of the polymer-based layer, and wherein, in the heating, the polymer-based layer is heated to a temperature above the melting temperature of the polymer;

optionally wherein, after the heating, the polymer-based layer is pre-cooled to a forming temperature before being stacked.

29. The method according to claim 28, wherein the metal layer is pre-heat treated to a forming temperature before being stacked.

30. The method according to claim 29, wherein the metal layer is first heated to an elevated temperature higher than the forming temperature, and then is rapidly pre-cooled to the forming temperature before being stacked.

31. The method according to claim 29 or claim 30, wherein the metal layer comprises a hot-formable alloy.

32. The method according to claim 28, wherein the metal layer is not heated before being stacked.

33. The method according to any preceding claim, wherein heating of the polymer-based layer and/or the metal layer before being stacked comprises furnace heating, radiation heating, convection heating, infrared heating, or electromagnetic induction heating.

34. The method according to any preceding claim, wherein the tool is maintained at a temperature lower than the glass transition temperature of the polymer of the polymer-based layer.

35. The method according to any preceding claim, wherein the multi-material structure is cooled in the tool until it reaches a temperature lower than the glass transition temperature of its constituent polymer(s).

36. The method according to claim 35, wherein the multi-material structure is cooled in the tool for less than one minute, preferably less than 30 seconds, preferably around 10 seconds.

37. The method according to any of claims 34 to 36, further comprising applying thermal treating to the tool.

38. The method according to claim 37, wherein the thermal treating is controlled using active cooling systems and/or heating elements.

39. The method according to any of claims 35 to 38, further comprising quenching the multi-material structure in the tool.

40. The method according to any preceding claim, further comprising performing a post-form heat treatment on the multi-material structure.

41. The method according to any preceding claim, wherein the multi-material structure comprises a plurality of polymer-based layers and/or a plurality of metal layers that are bonded by the method.

42. A multi-material structure manufactured by the method of any preceding claim.

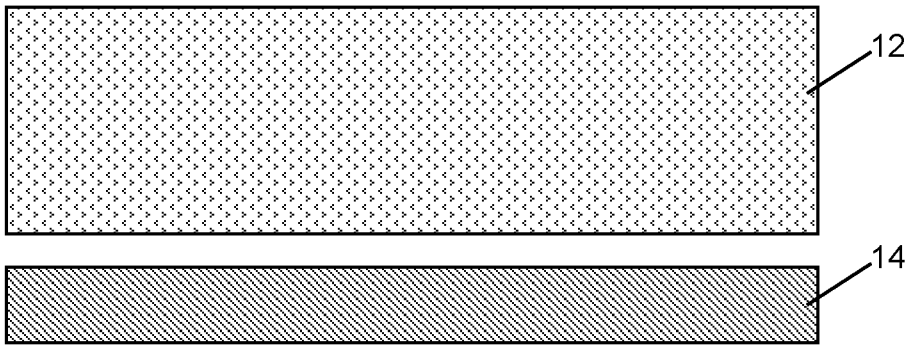


Figure 1

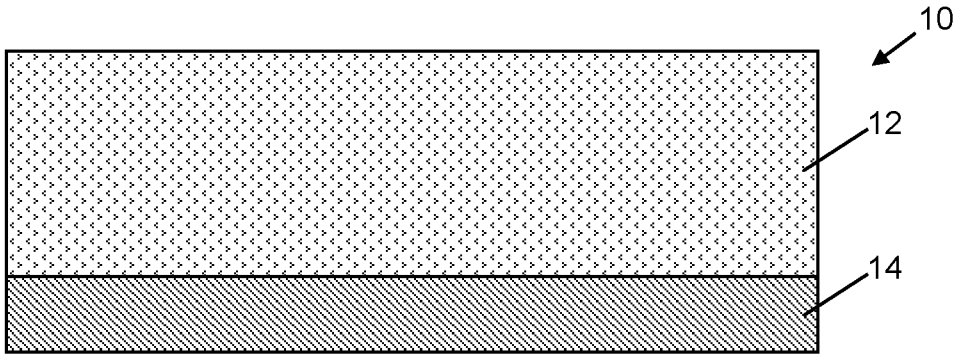


Figure 2

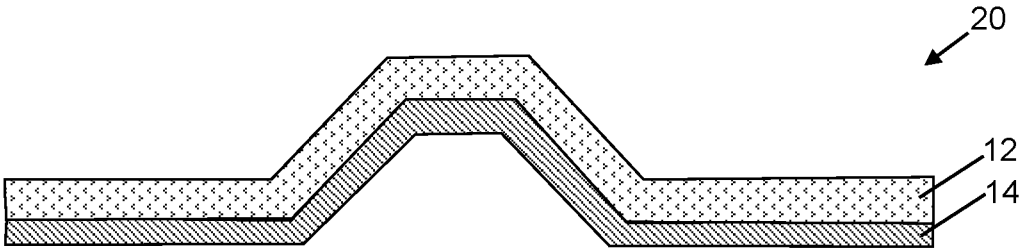


Figure 3

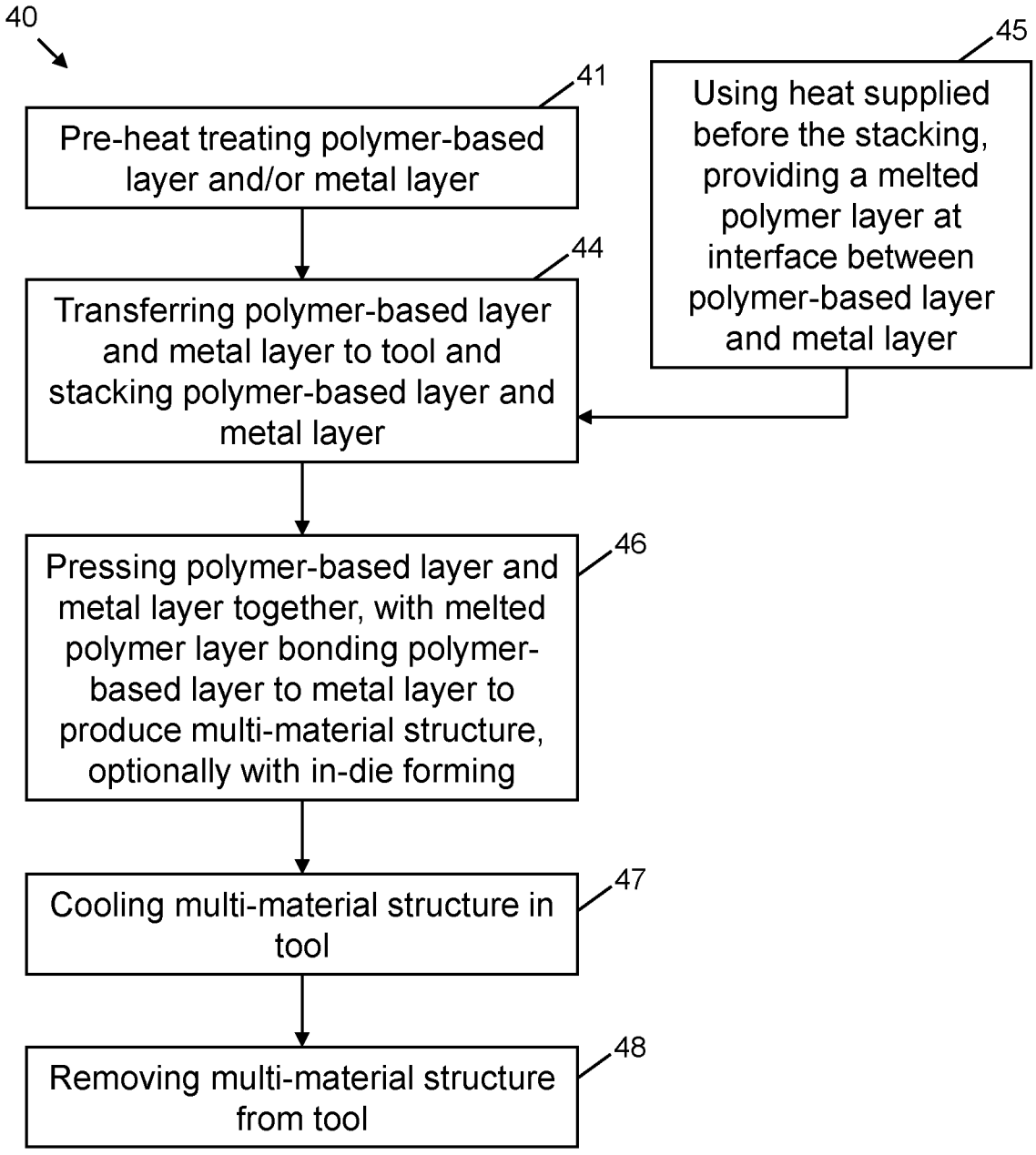

40
41
Pre-heat treating polymer-based layer and/or metal layer
44
Transferring polymer-based layer and metal layer to tool and stacking polymer-based layer and metal layer
45
Using heat supplied before the stacking, providing a melted polymer layer at interface between polymer-based layer and metal layer
46
Pressing polymer-based layer and metal layer together, with melted polymer layer bonding polymer-based layer to metal layer to produce multi-material structure, optionally with in-die forming
47
Cooling multi-material structure in tool
48
Removing multi-material structure from tool

Figure 4

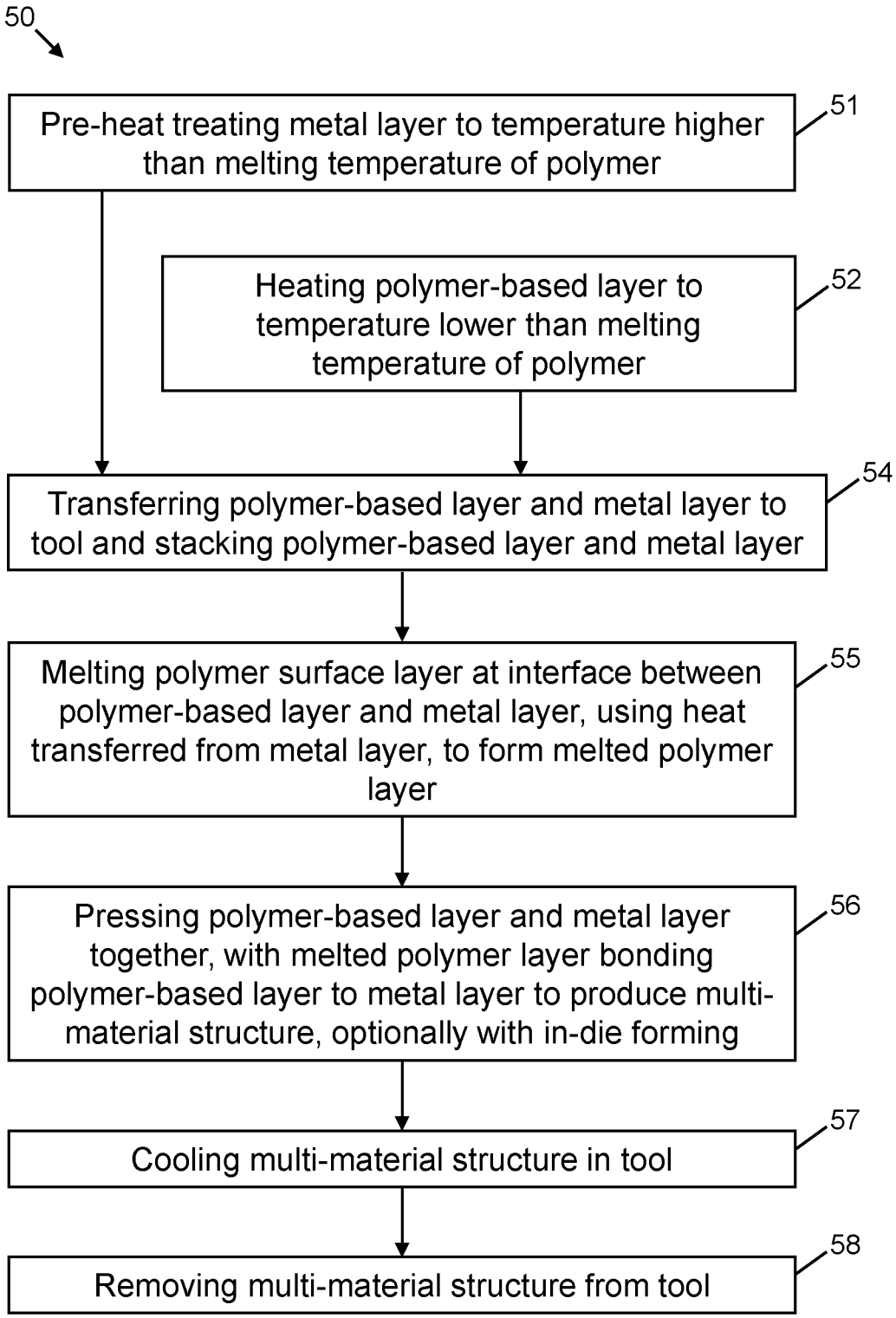

50
Pre-heat treating metal layer to temperature higher than melting temperature of polymer
51
Heating polymer-based layer to temperature lower than melting temperature of polymer
52
Transferring polymer-based layer and metal layer to tool and stacking polymer-based layer and metal layer
54
Melting polymer surface layer at interface between polymer-based layer and metal layer, using heat transferred from metal layer, to form melted polymer layer
55
Pressing polymer-based layer and metal layer together, with melted polymer layer bonding polymer-based layer to metal layer to produce multi-material structure, optionally with in-die forming
56
Cooling multi-material structure in tool
57
Removing multi-material structure from tool
58

Figure 5

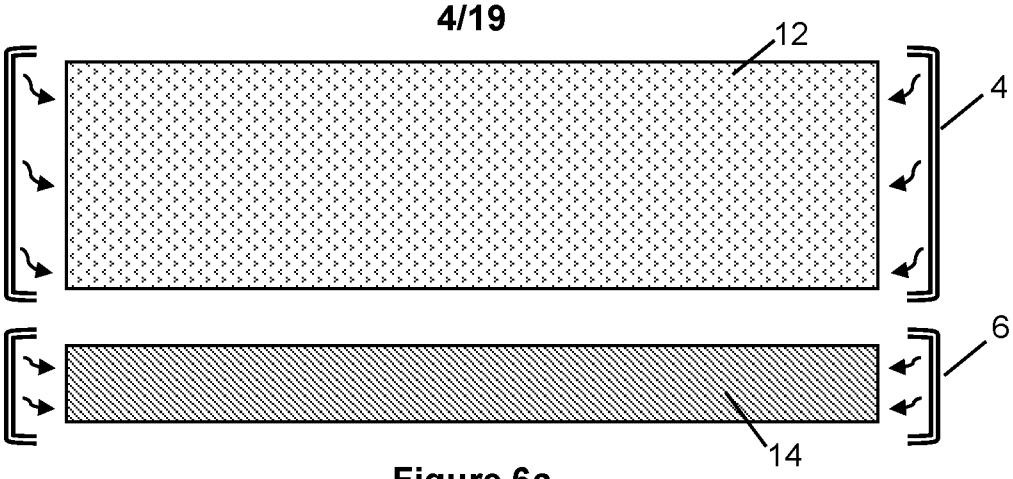


Figure 6a

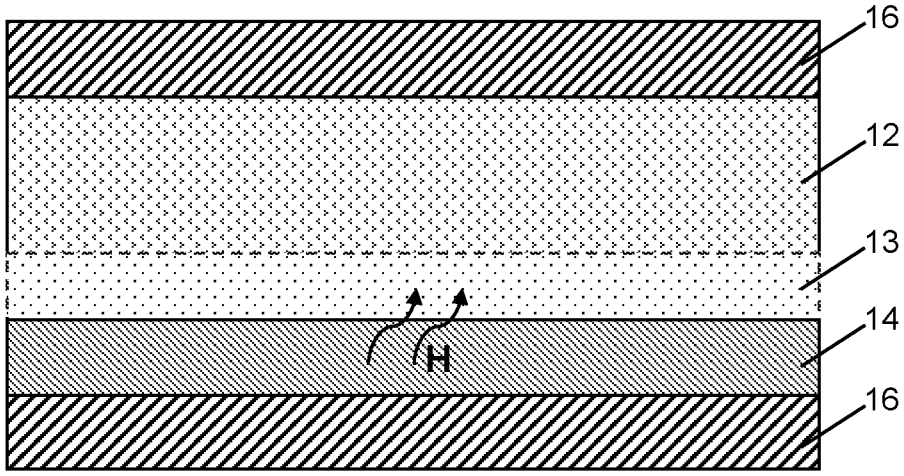


Figure 6b

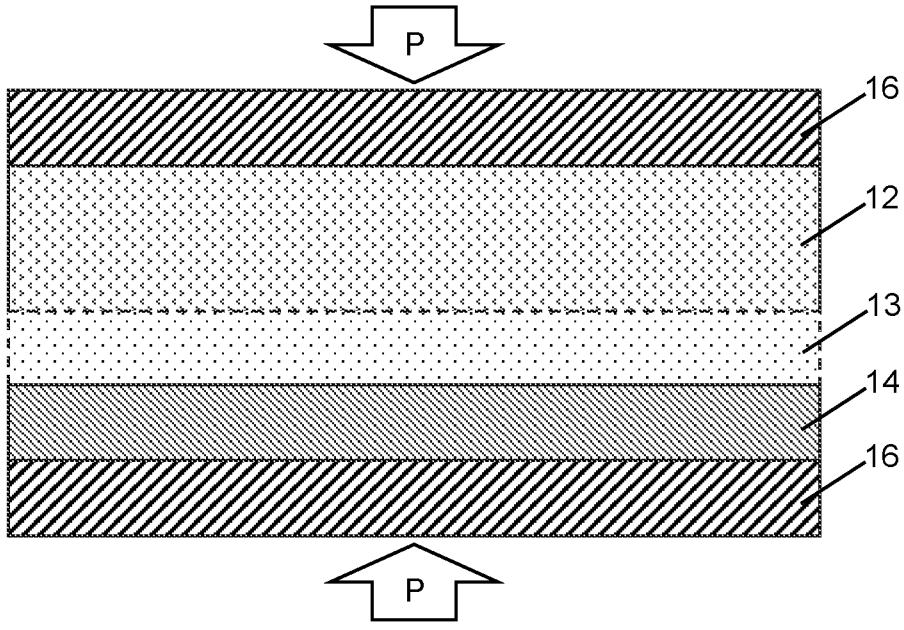


Figure 6c

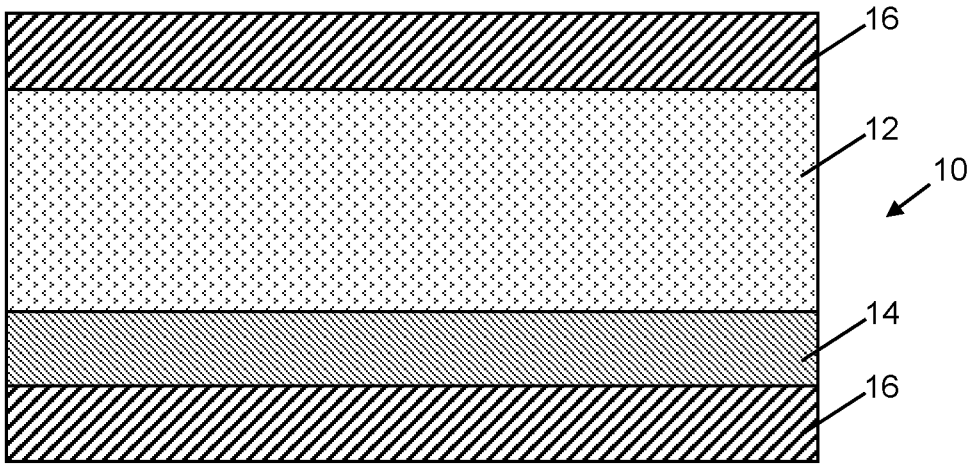


Figure 6d

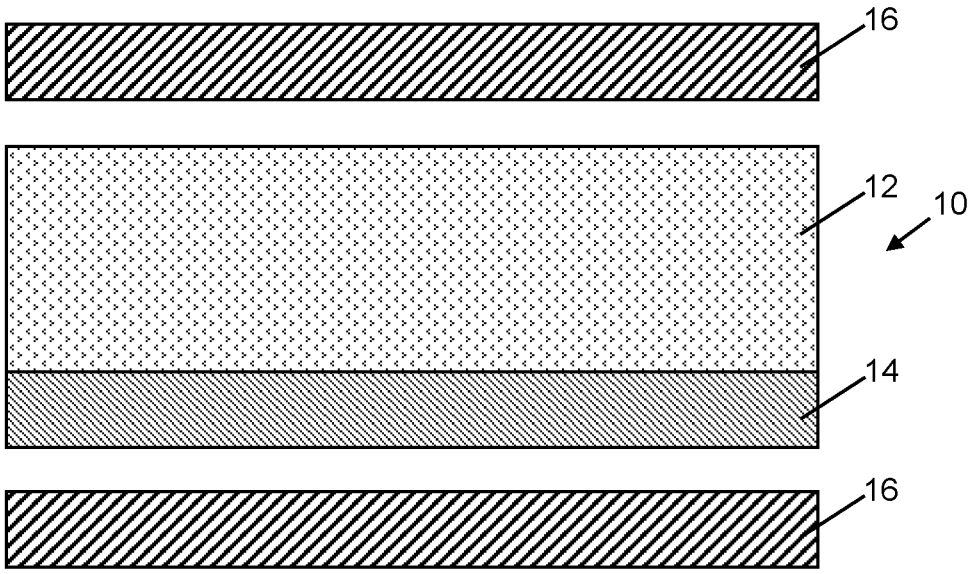


Figure 6e

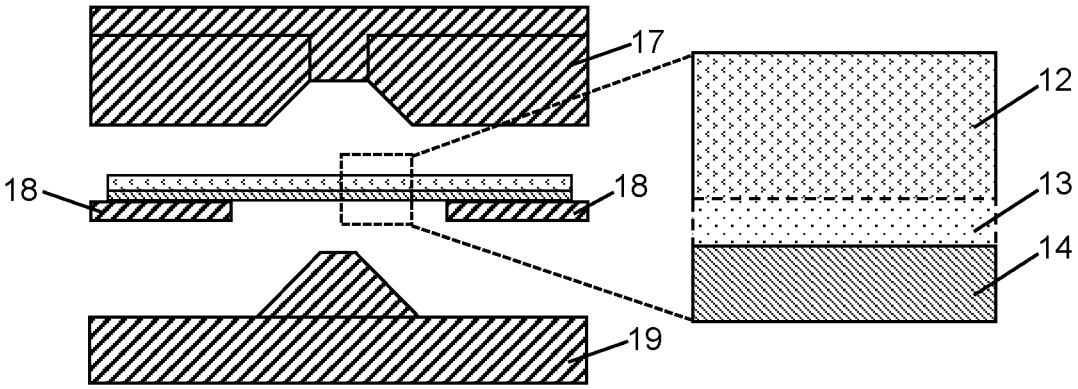


Figure 7a

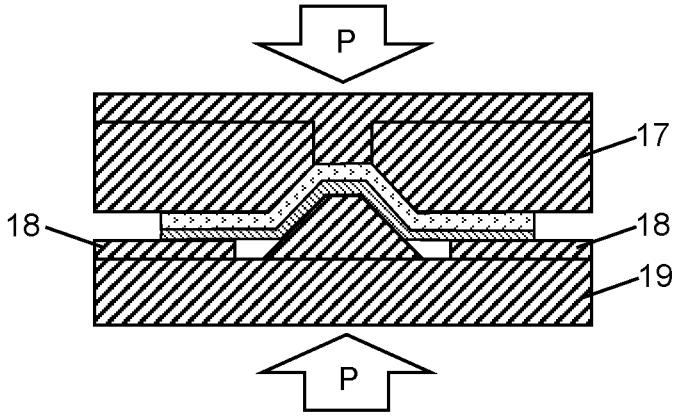


Figure 7b

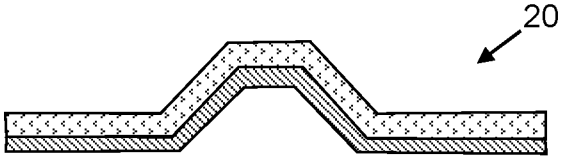


Figure 7c

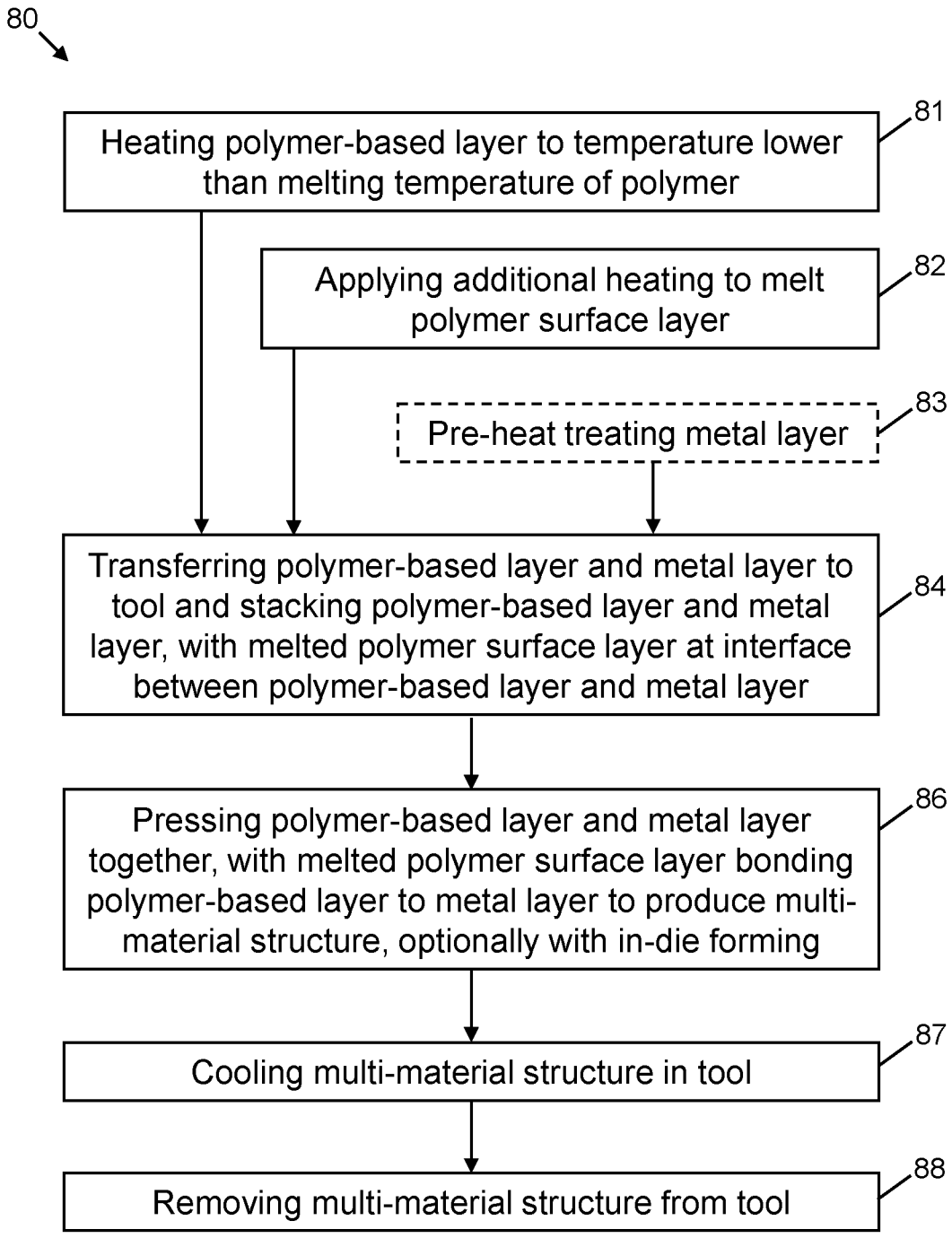

80
Heating polymer-based layer to temperature lower than melting temperature of polymer
81
Applying additional heating to melt polymer surface layer
82
Pre-heat treating metal layer
83
Transferring polymer-based layer and metal layer to tool and stacking polymer-based layer and metal layer, with melted polymer surface layer at interface between polymer-based layer and metal layer
84
Pressing polymer-based layer and metal layer together, with melted polymer surface layer bonding polymer-based layer to metal layer to produce multi-material structure, optionally with in-die forming
86
Cooling multi-material structure in tool
87
Removing multi-material structure from tool
88

Figure 8

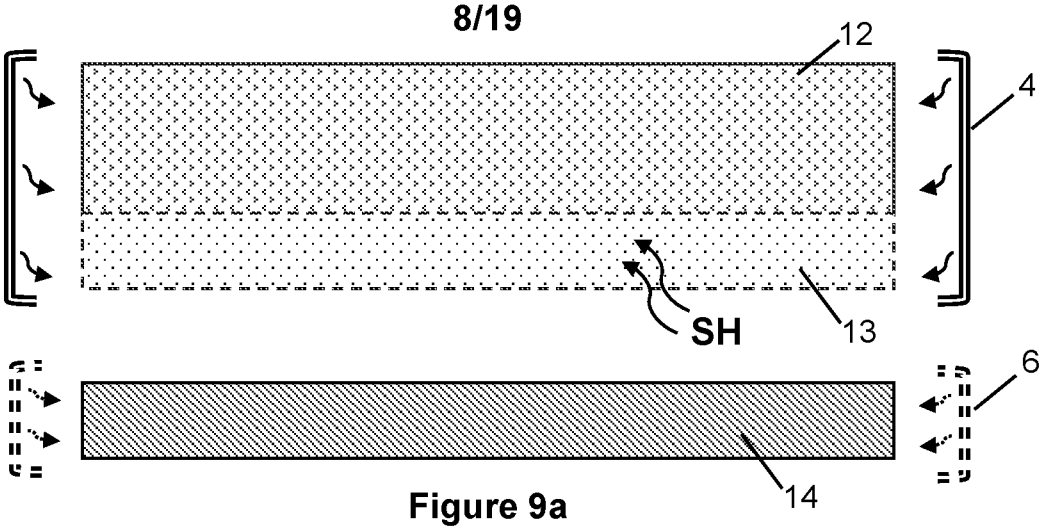


Figure 9a

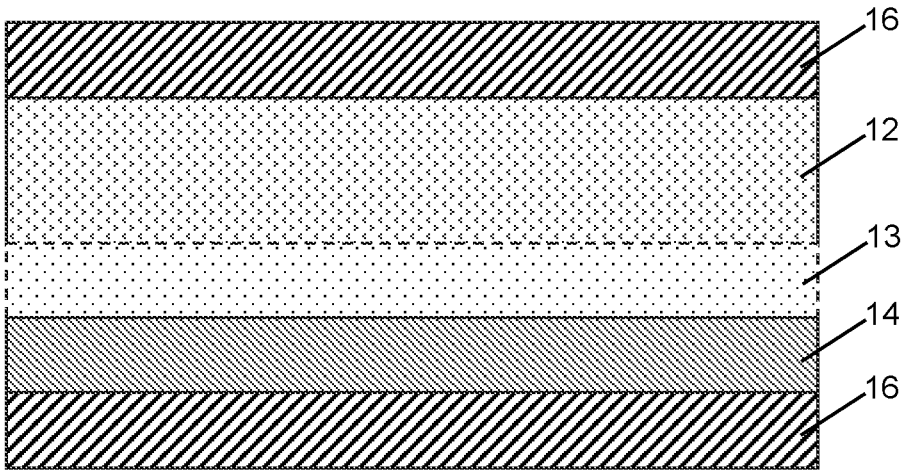


Figure 9b

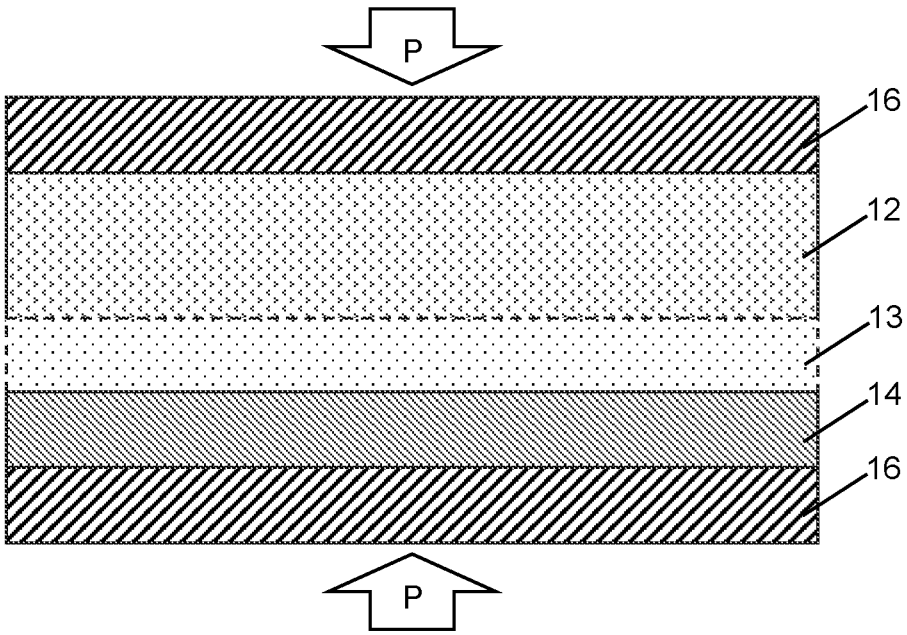


Figure 9c

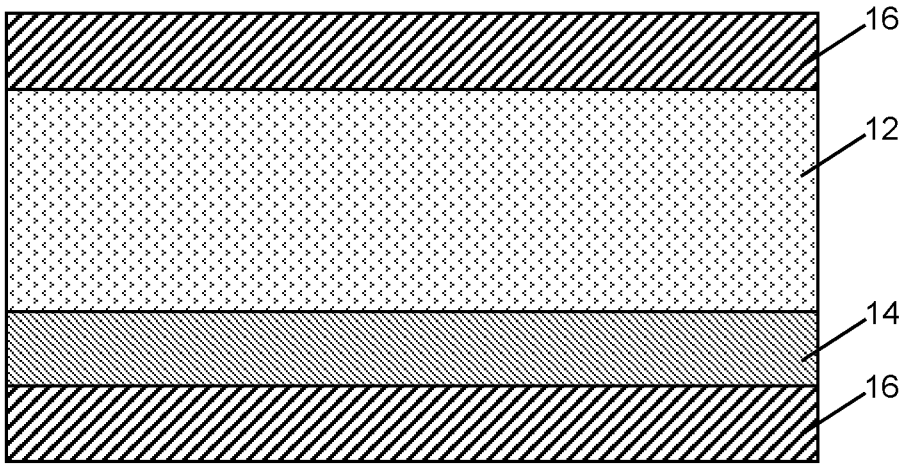


Figure 9d

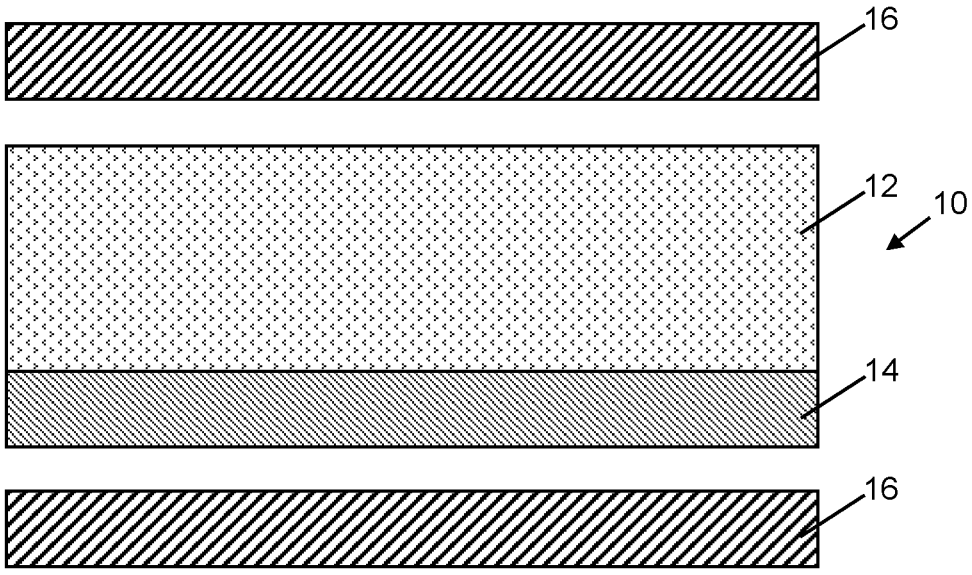


Figure 9e

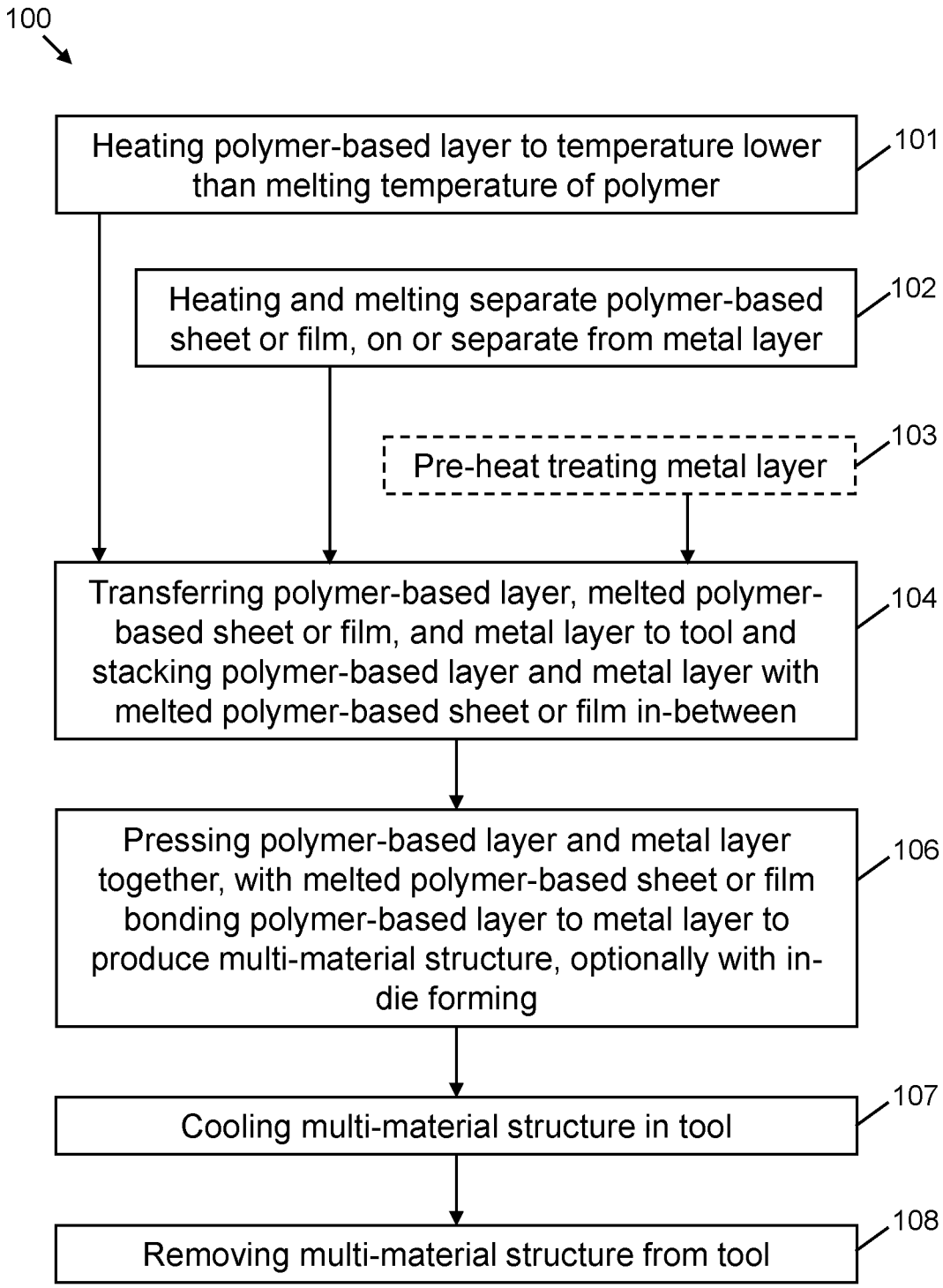

100
Heating polymer-based layer to temperature lower than melting temperature of polymer
101
Heating and melting separate polymer-based sheet or film, on or separate from metal layer
102
Pre-heat treating metal layer
103
Transferring polymer-based layer, melted polymer-based sheet or film, and metal layer to tool and stacking polymer-based layer and metal layer with melted polymer-based sheet or film in-between
104
Pressing polymer-based layer and metal layer together, with melted polymer-based sheet or film bonding polymer-based layer to metal layer to produce multi-material structure, optionally with in-die forming
106
Cooling multi-material structure in tool
107
Removing multi-material structure from tool
108

Figure 10

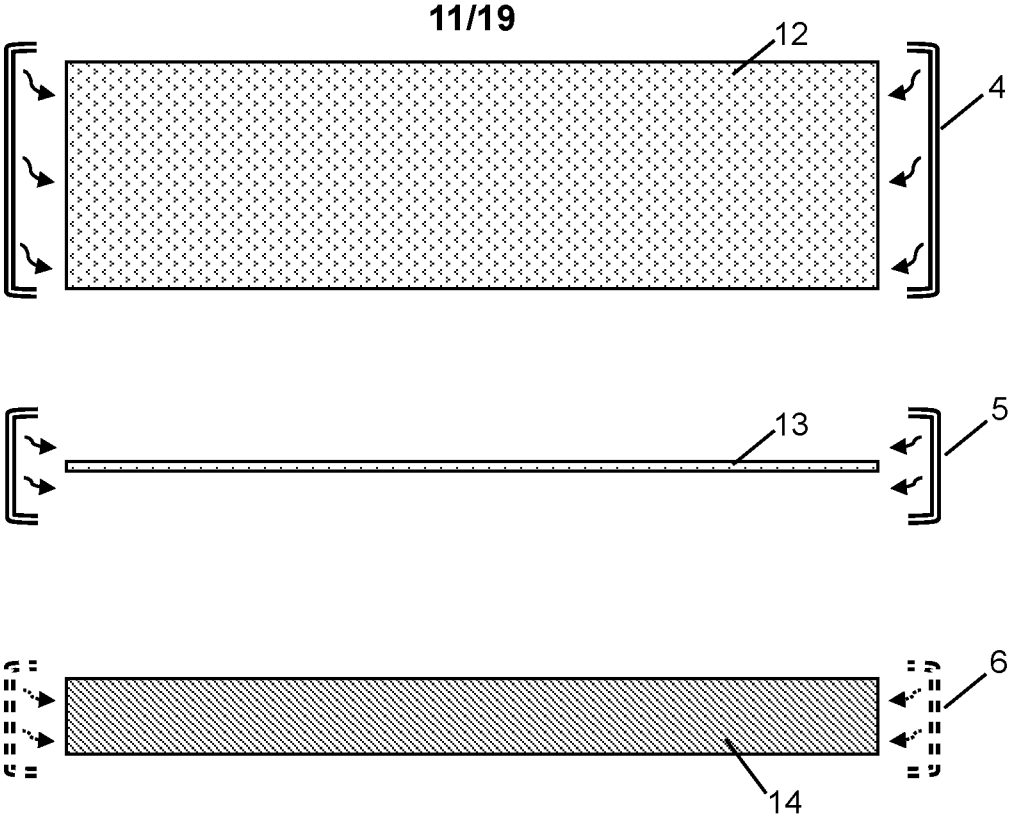


Figure 11a

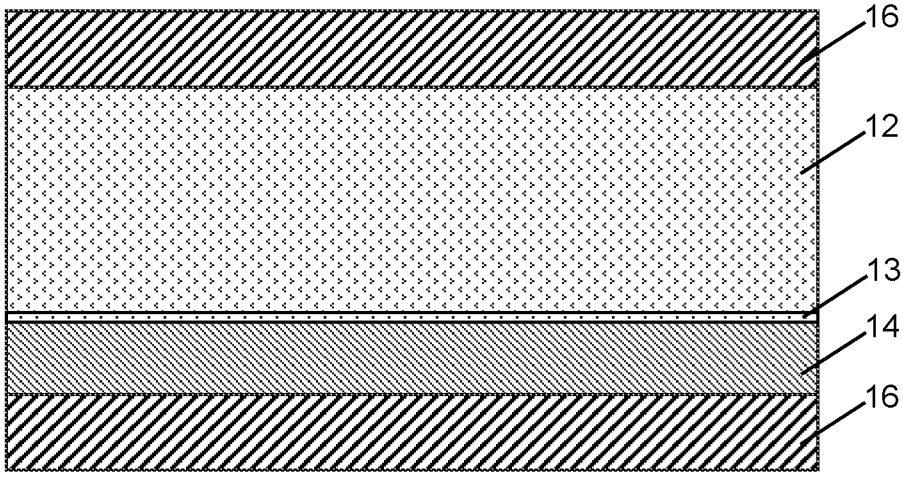


Figure 11b

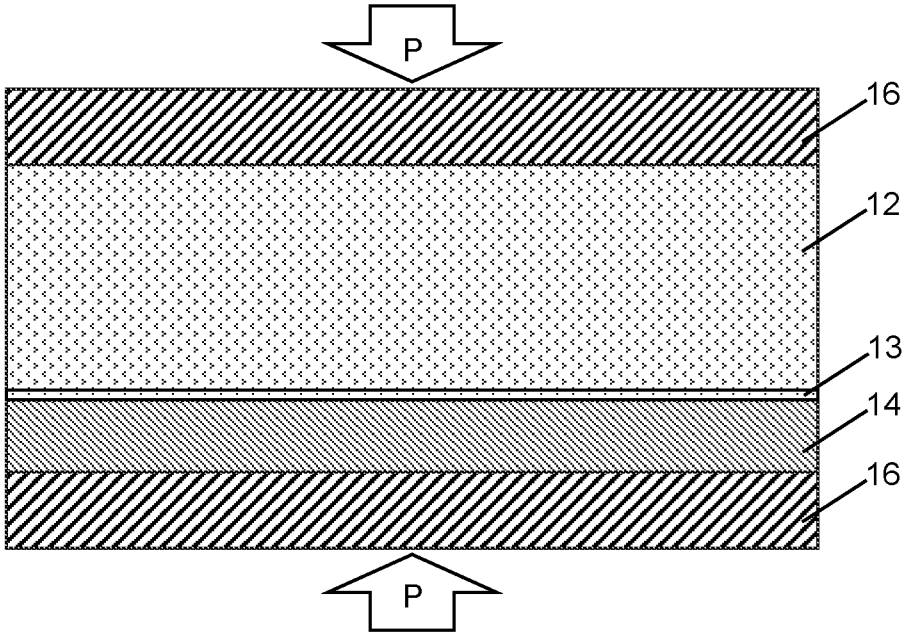


Figure 11c

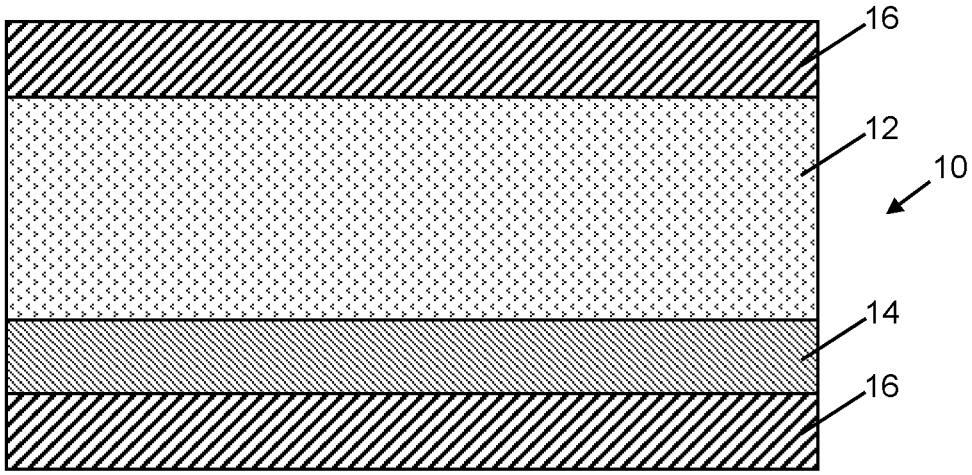


Figure 11d

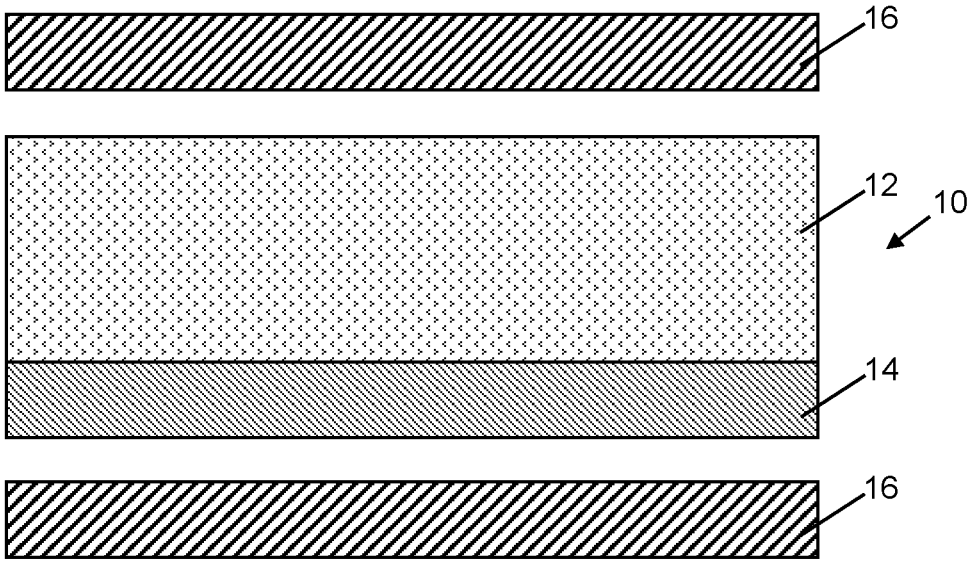


Figure 11e

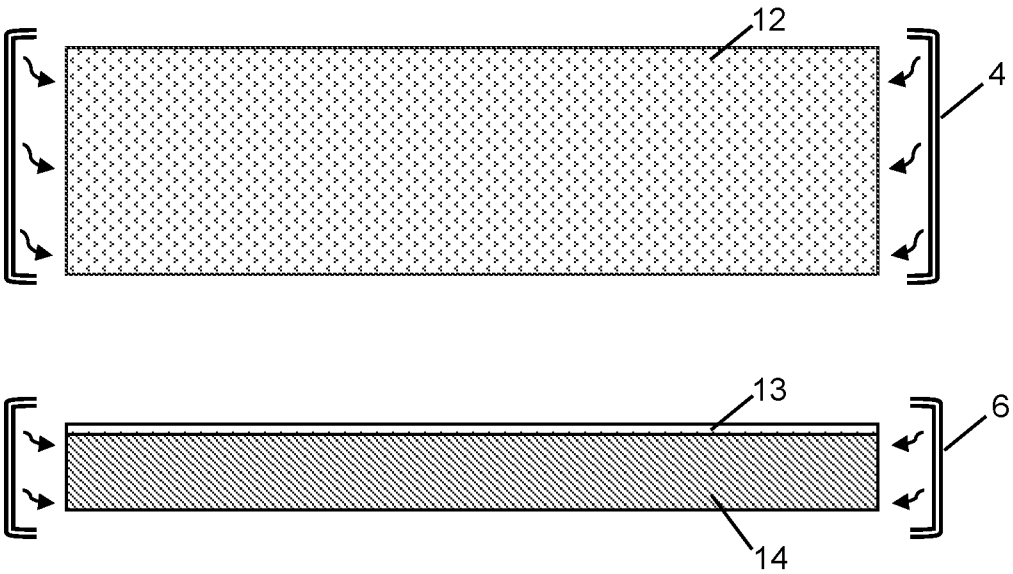


Figure 11f

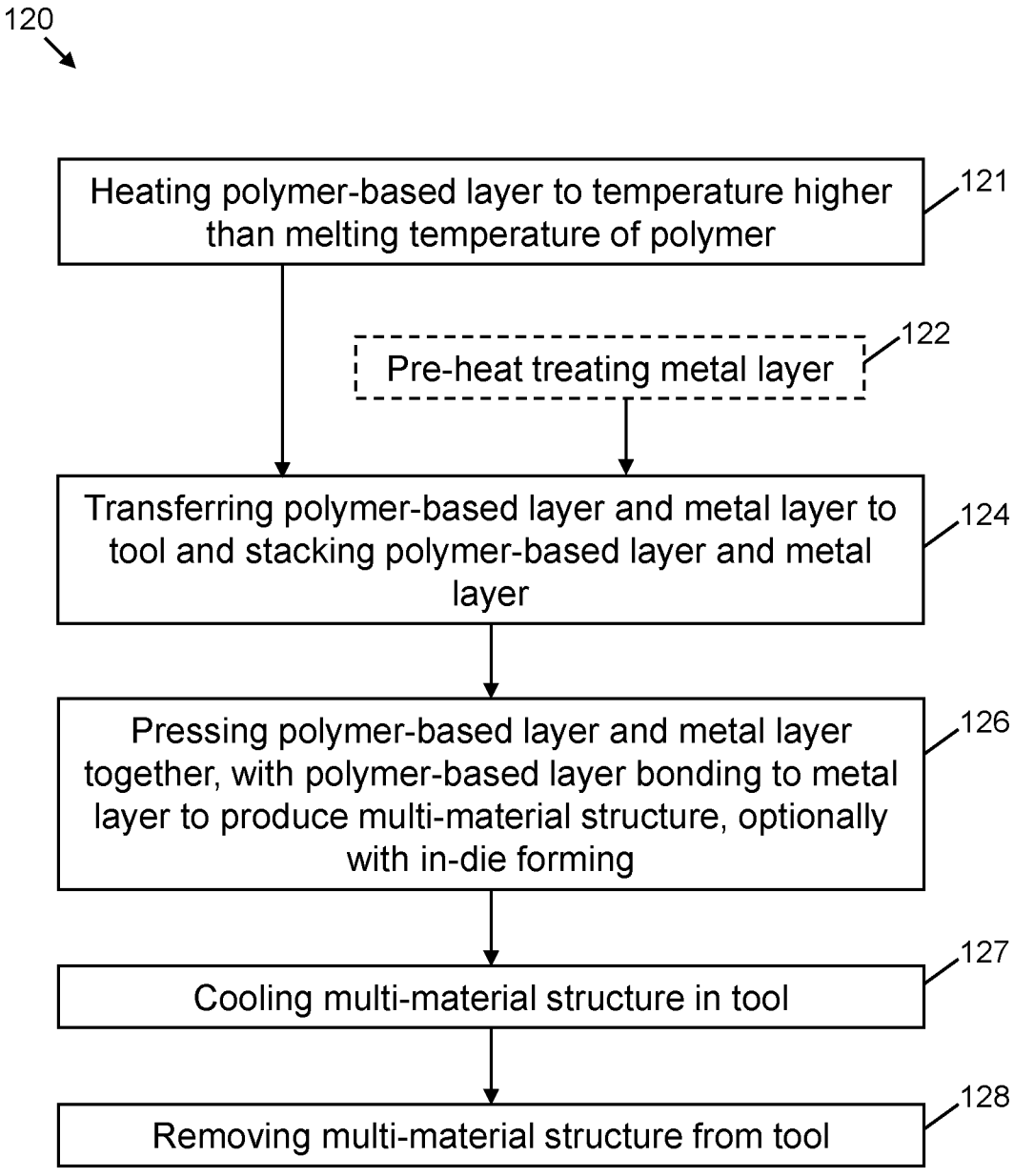

120
Heating polymer-based layer to temperature higher than melting temperature of polymer
121
Pre-heat treating metal layer
122
Transferring polymer-based layer and metal layer to tool and stacking polymer-based layer and metal layer
124
Pressing polymer-based layer and metal layer together, with polymer-based layer bonding to metal layer to produce multi-material structure, optionally with in-die forming
126
Cooling multi-material structure in tool
127
Removing multi-material structure from tool
128

Figure 12

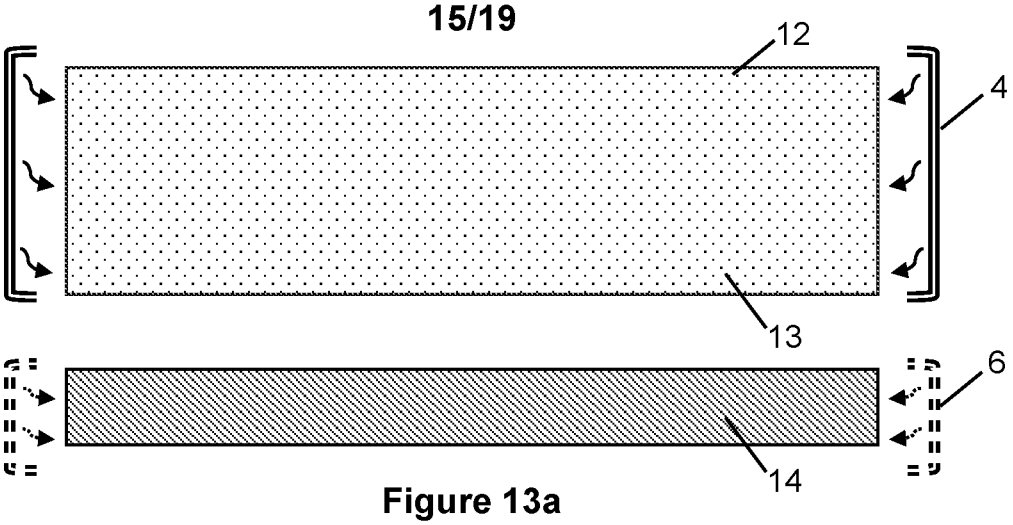


Figure 13a

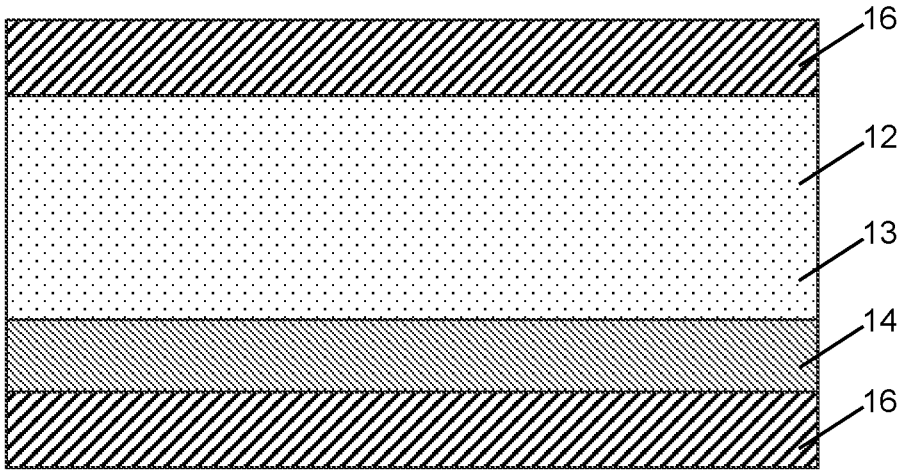


Figure 13b

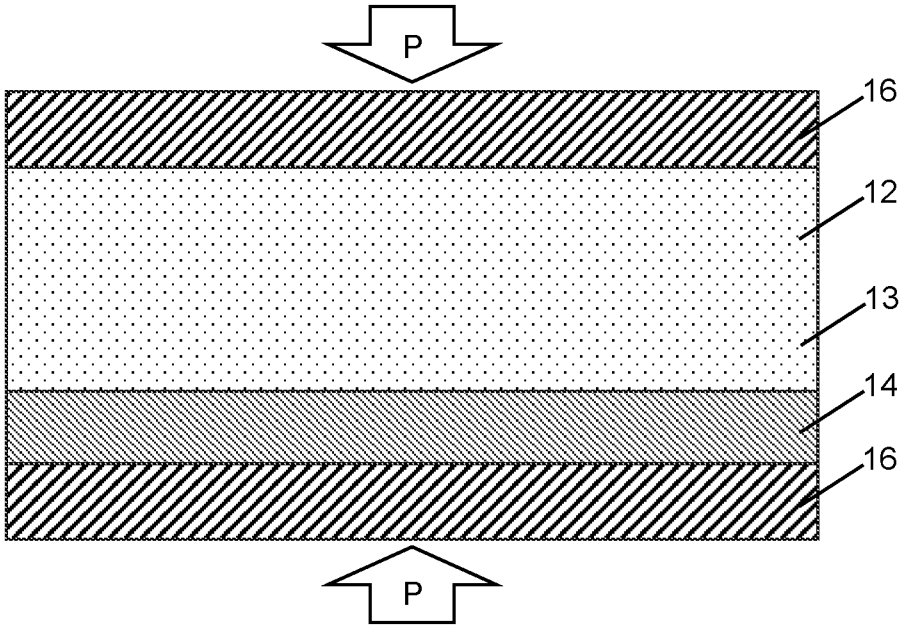


Figure 13c

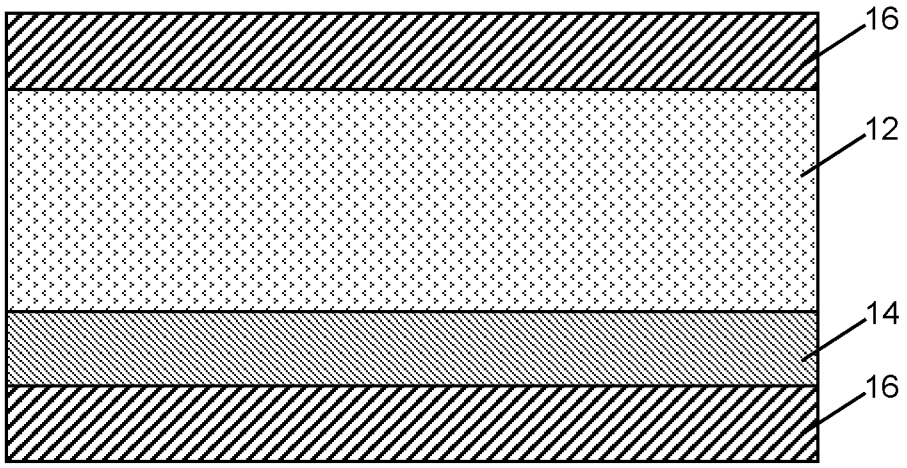


Figure 13d

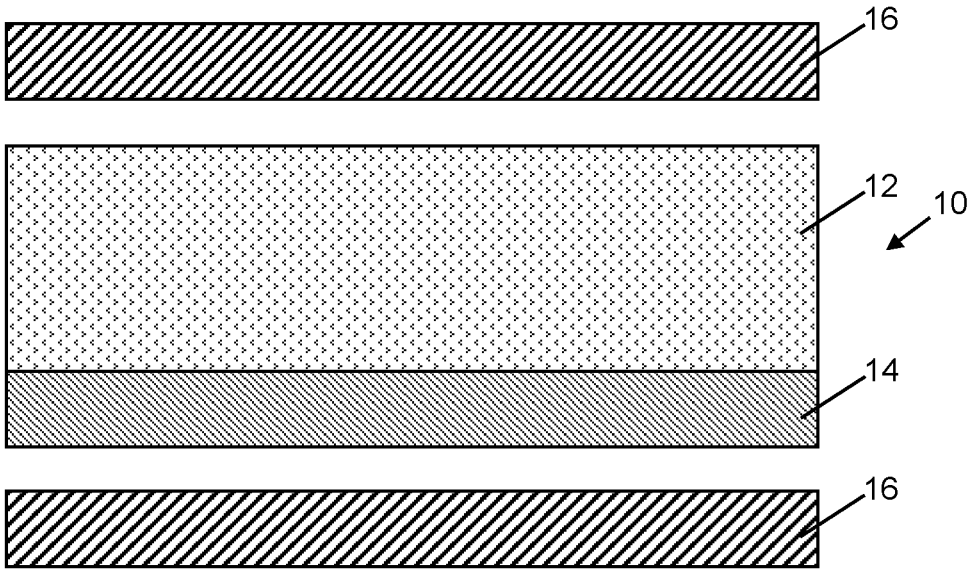


Figure 13e

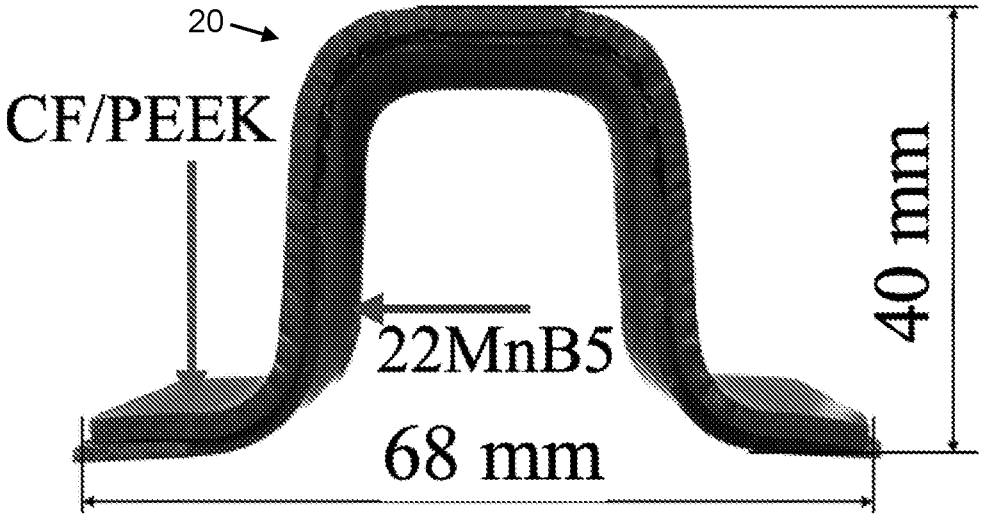


Figure 14

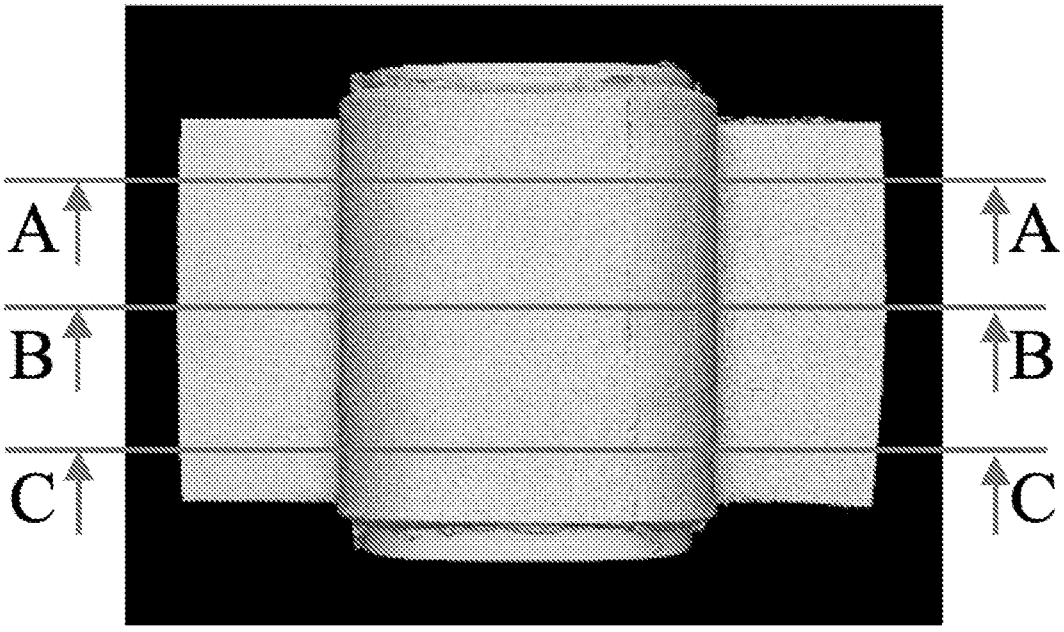


Figure 15

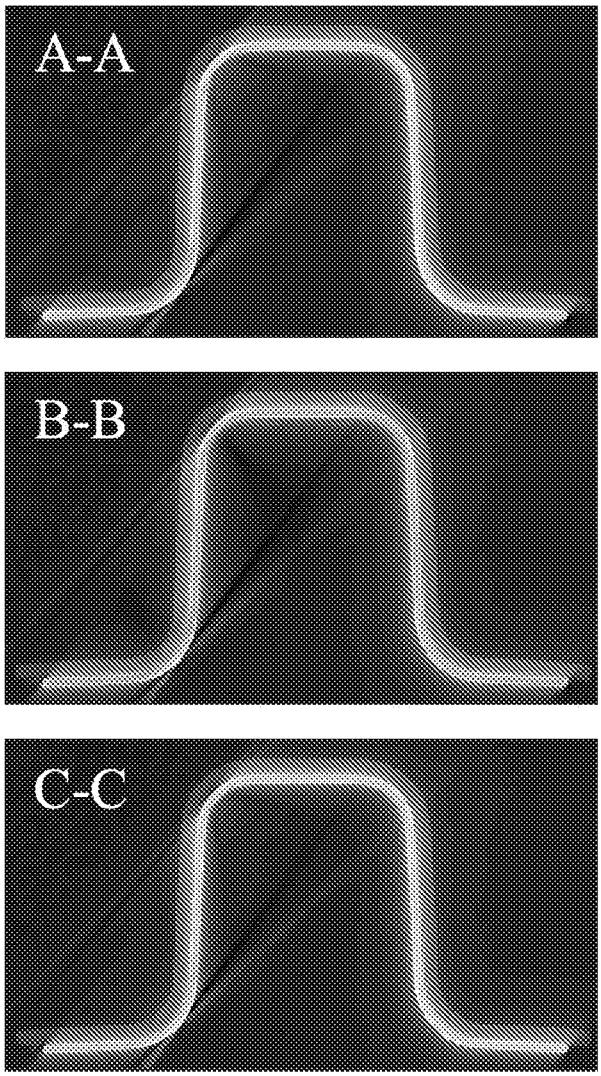


Figure 16

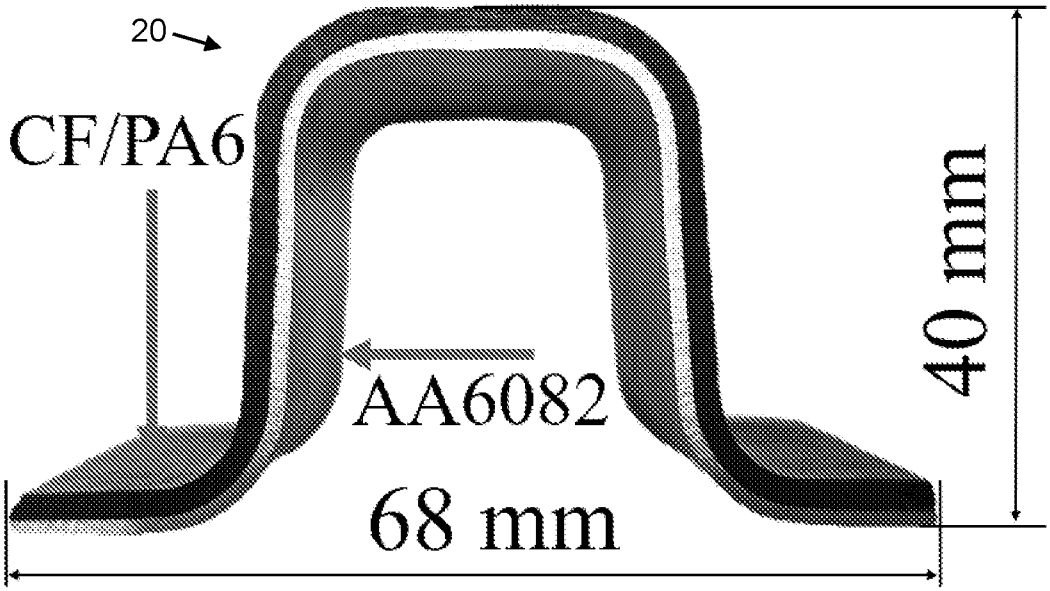


Figure 17

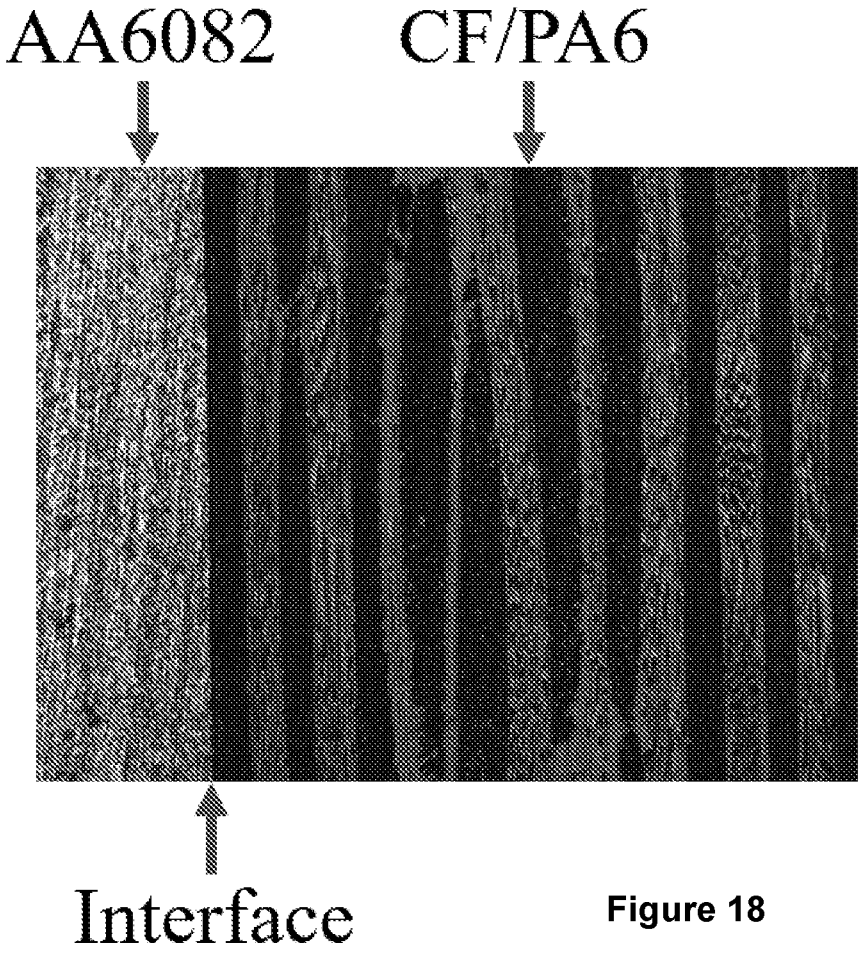


Figure 18

INTERNATIONAL SEARCH REPORT

International application No
PCT/GB2025/052703

A. CLASSIFICATION OF SUBJECT MATTER
INV. B32B37/04 B32B37/06 B32B37/08
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
B32B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 11 135 825 B2 (NIPPON STEEL CHEMICAL & MAT CO LTD [JP]) 5 October 2021 (2021-10-05) column 7, line 26 - column 24, line 12 figures 1-4	1-42
X	GB 2 610 024 A (UNIV CENTRAL SOUTH; UNIV CENTRAL SOUTH [CN]) 22 February 2023 (2023-02-22) paragraphs [0044] - [0050] figure 1	42
X	US 10 829 193 B2 (PACCHIONE MARCO [DE]; FIDEU PAULIN [DE] ET AL.) 10 November 2020 (2020-11-10) column 16, line 27 - column 18, line 58 figure 1	42

-/--

[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
17 March 2026	31/03/2026

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Bataille, Laurent

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Form PCT/ISA/210 (second sheet) (April 2005)

INTERNATIONAL SEARCH REPORT

International application No
PCT/GB2025/052703

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	EP 3 072 683 B1 (PLASTIC OMNIUM CIE [FR]) 4 November 2020 (2020-11-04) paragraphs [0019], [0020] claim 1 figures 3-5 -----	42

3

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/GB2025/052703

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
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		GB 2610024 A	22-02-2023
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		EP 3072683 A1	28-09-2016
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		WO 2015075683 A1	28-05-2015